

OpenAI's 719 AI-generated math papers, explained from zero

On 2026-10-06 OpenAI published **722 math manuscripts (372 result families)** generated by an unreleased internal model. Among the claims: the quasi-Riemann hypothesis, the Unique Games Conjecture, matrix-multiplication exponent $\omega \leq 9/4$, non-amenability of Thompson's group F , and the plane not being 5-colorable — a run of problems open for decades. The next day three manuscripts were withdrawn over a sign error. Mathematicians dubbed it the "Mathocalypse." The goal of this deck: using **only linear algebra, calculus and high-school math**, take 10 representative results far enough that you can judge how big each is, how hard it was, why it matters, and how far it has been verified.

First edition 2026-10-08 (Part II outlook §17–§22 added the same day) · repo snapshot openai/math@fd4aeeb (includes the Oct 7 withdrawals and fixes) · every fact is sourced; [P] = primary material read (paper / repo / Lean source), [R] = secondary reporting, Δ = unverified or conflicting · English edition of notes.html (Chinese)

ORIENTATION

0. What this deck will and won't give you

YOU WILL GET

A "ladder" understanding of each of 10 headline results: start from a toy example you can compute by hand and climb level by level to the real statement; learn how long the problem was stuck and where; see exactly what OpenAI claims, how much the Lean formalization covers, and what human experts have said so far; and what it means for mathematics, for AI, and for AI safety.

YOU WON'T GET

The ability to judge whether any proof is correct. As of 2026-10-08, **none** of these 10 results has a human domain expert publicly saying "I have read and checked it."

"Verification" in this deck means two things only: (1) a static review of the Lean source and statements, and (2) a summary of public reactions. Both are labeled with sources and confidence.

0.1 Reading path

§1 fills in three basics (what mathematicians do, what an "open problem" is, how to judge how big a result is). §2–§3 cover the event and what Lean actually guarantees — **these two**

sections determine how you should read every "claim" that follows. §4–§14 are the 10 results, each with the same structure: *ladder* → *problem* → *why it was hard* → *what OpenAI claims* → *verification status* → *if true*; read them in any order. §15–§16 give cross-cutting patterns and implications for AI. **Part II (§17–§22) looks ahead**: three trend curves (task length, open-problem solve rate, formal-proof cost), why bottlenecks move from proofs to statements and specs, what AI math and coding imply for debate, singular learning theory, Guaranteed Safe AI and neurosymbolic systems, and a T0–T4 timeline with falsifiers. §23 collects reactions, §24 lists open questions and a learning path, §25 is the glossary and sources.

0.2 Cheat sheet

§	RESULT (FAMILY NO.)	OPEN FOR	LEAN	ONE LINE
4	Quasi-Riemann hypothesis (003)	~125 yrs	yes	ζ has no zeros at all in $\text{Re } s > 7/8$
5	Irrationality exponent of $\pi = 2$ (017)	~70 yrs	yes $\triangle$	π cannot be approximated "unusually well" by fractions
6	Matrix multiplication $\omega \leq 9/4$ (107)	57 yrs	yes	a drop of 0.12 at once; the previous 12 years gave 0.0017
7	Unique Games Conjecture (102)	24 yrs	yes	optimality of many approximation algorithms becomes theorem
8	$L = RL = BPL$ (103)	~30 yrs	yes	randomness is useless for small-memory computation
9	The plane is not 5-colorable (158)	76 yrs (last lower-bound step 2018)	yes	chromatic number of the plane $\in \{6, 7\}$
10	Thompson's group F non-amenable (248)	47 yrs	yes	famous for "proofs" in both directions later withdrawn
11	Navier–Stokes blowup (separate Sept release)	Millennium Prize problem	yes	only the "with forcing" alternative; credit disputed
12	Relativistic Vlasov–Maxwell (362)	40 yrs	yes	large-data global smoothness for a plasma model (one species)
13	Mézard–Parisi formula (221)	25 yrs	yes	free-energy formula for sparse spin glasses (even-arity case)
14	Hodge-related withdrawals (032 etc.)	—	no	one sign error, three papers withdrawn: the risk of no formalization

1. What do mathematicians actually do?

In one sentence: **they study the structure of abstract objects and establish truths about them by proof.** What separates mathematics from science and engineering is *how you know something is right*: mathematics by logical proof (once proven, true forever), science by experiment (can be overturned), engineering by "it works and measures well enough."

1.1 Four words

WORD	MEANING	EXAMPLE FROM THIS EVENT
conjecture	a statement widely believed but unproven	the Unique Games Conjecture (Khot 2002)
theorem	a statement that has been proven	the Goemans–Williamson 0.878 algorithm
counterexample	a concrete object showing that a "for all ..." conjecture is false	the 9-dimensional Borsuk counterexample; about 20% of this batch are counterexamples/disproofs
formalization	writing a proof as code a computer can check step by step (here Lean 4)	about 300/719 top-line results have Lean proofs

1.2 Two cultures of mathematics

In *The Two Cultures of Mathematics* (2000), Fields medalist Tim Gowers splits mathematicians into **problem-solvers** (who crack specific hard problems, like Erdős problems) and **theory-builders** (who invent new concepts and frameworks, like "group" or "manifold"). Thurston's *On Proof and Progress in Mathematics* (1994) goes further: the point of mathematics is not to produce theorems but to **advance human understanding**. These two essays are the key to this controversy — what AI has broken open at scale is the problem-solving culture, and that is what Tao means by "the end of Math 1.0."

1.3 Five questions for judging how big a result is

1. **How long was it open, and who tried?** A problem open for 50 years with repeated failures by top experts weighs very differently from one posed two years ago.
2. **Last step or giant leap?** Moving ω from 2.3713 to 2.3711 is the former; to 2.25 is the latter.

3. **How much does it unlock?** If UGC holds, hundreds of "if UGC then ..." papers become theorems at once.
4. **Is the method new?** New methods often matter more than the result, because they can be reused.
5. **Who chose the problem, and how many attempts failed?** The most overlooked question in this event; see §2.3 and §15.

THE EVENT · TIMELINE AND NUMBERS

2. What happened

DATE	EVENT
2025–07	An OpenAI experimental model and Google DeepMind's Gemini Deep Think reach gold-medal scores at IMO 2025 (35/42) — competition problems with known answers fall. [R]
2026–09–08	OpenAI announces that ~10,000 agents over 88 hours proved finite-time blowup "with forcing" for 3D Navier–Stokes (166 pages + Lean), sparking a credit dispute (§11). [P] [R]
late 2026–09	The Advisory Group on Mathematics and AI hosted by the Institute for Advanced Study (members include Terence Tao) publishes recommendations for releasing AI-generated mathematics. [R]
2026–10–06	openai/math released: 722 manuscripts / 372 result families, Apache-2.0, with a Lean library and 10 abridged reasoning summaries. [P]
2026–10–07	3 manuscripts withdrawn (one sign error, cascading), 14 repaired, 13 re-cited, 6 new formalizations → 719 manuscripts , "300/719 $\approx$ 42% of top-line results formalized." A statement from the Association for Human Mathematics (AHM) is reposted on Tao's blog; Scott Aaronson posts "The Mathocalypse". [P] [R]
2026–10–08	Tao's blog runs a guest post by Lozano-Robledo, What should we tell our students? [P]

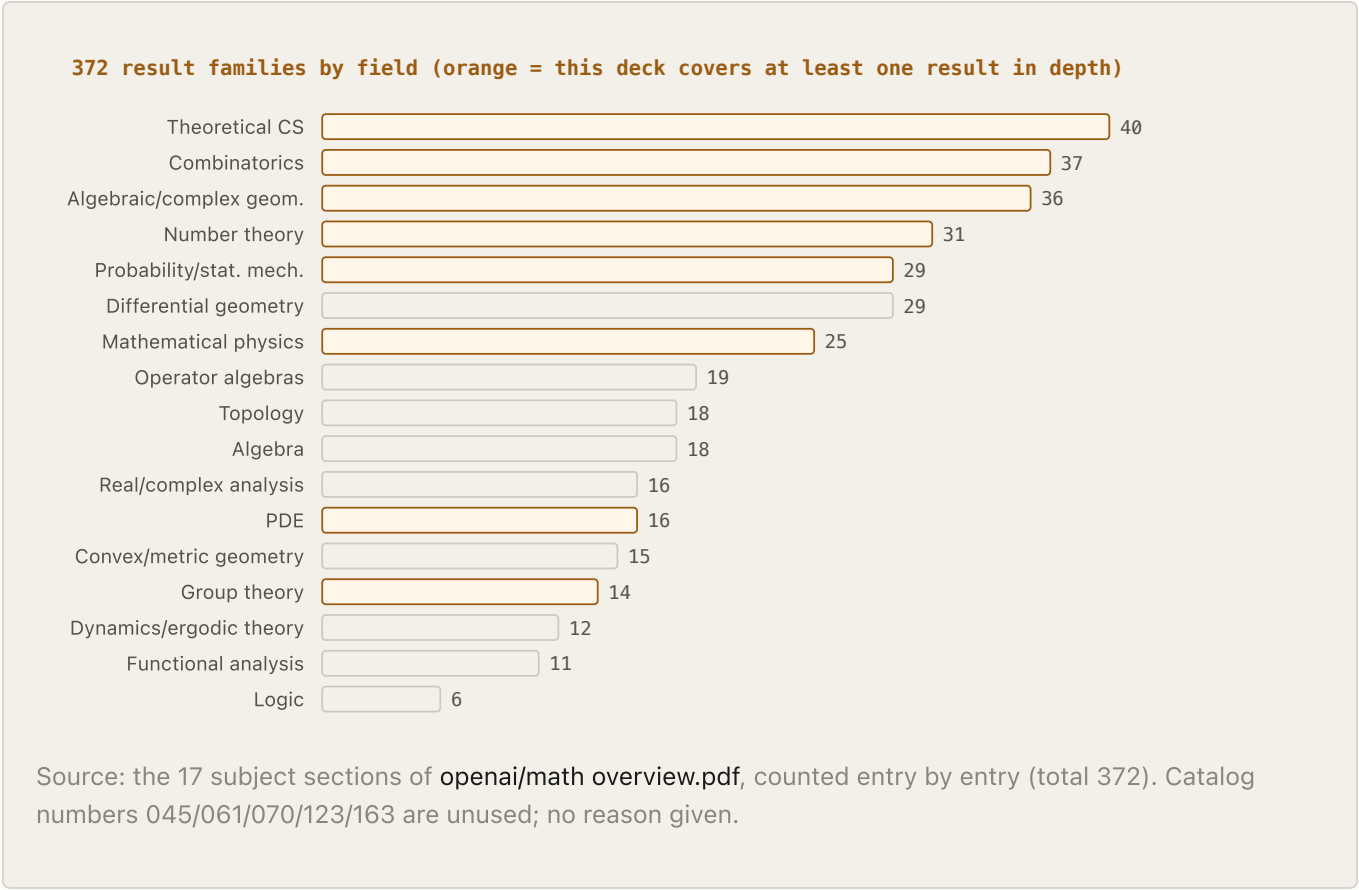
2.1 How the results were produced

From the README [P]: "The vast majority of results were obtained with the same procedure using an unreleased internal OpenAI model. On average, each result used three hours of ChatGPT Pro thinking compute... the model was posed approximately 4,000 problems... requiring an appropriate level of significance led to the catalog." In short: **~4,000 open problems → filtered for significance → 372 families**. An OpenAI spokesperson told Scientific American that almost everything came from "a single prompt handed to a single AI agent" [R]; MIT's Andrew Sutherland responded that until the model can be replicated, "one-shotting" claims should be treated as unverified — "We should ask for receipts."

Exceptions: the README concedes that the zero-free region for ζ (§4) and the Hodge conjecture for CM abelian varieties (§14) were *not* produced by the fixed procedure, and that the Re $s > 11/12$ write-up was human-edited. What exactly differed (more compute? human guidance?) is not disclosed [△](#).

2.2 Reconciling the numbers

CLAIM	WHAT IT ACTUALLY MEANS
"722" vs "719"	722 at release; 719 after the three Oct 7 withdrawals. Early news used 722.
"372 families" vs "377"	The catalog is numbered 001–377 with 045/061/070/123/163 unused: 372 actual families.
"~42% formalized"	$300 \text{ top-line results} \div 719 \text{ manuscripts}$ — numerator and denominator in different units. By family, $242/372 \approx 65\%$ have a Lean scope document.
"162 fully + 73 partially formalized"	Reported by some outlets; source not found △ .



2.3 The one thing to keep in mind: selection effects

Epoch AI's FrontierMath Erdős (2026-09) took 68 Erdős problems **pre-registered** as open and required Lean proofs; GPT-6 Astra solved 2 ($\approx 3\%$), all other models 0. OpenAI's release

is **post-hoc selection**: of ~4,000 problems it kept the "significant" ones and published no failure list. A rough hit rate is $\leq 9\%$ (by family) — but problem difficulty is not comparable. **Both numbers are right; they answer different questions**: the first asks "given an arbitrary open problem, can it solve it?", the second "given 4,000, how many good results can it pick out?"

VERIFICATION · BEFORE READING ANY "CLAIM"

3. What "has a Lean proof" actually guarantees

Lean is a proof assistant: you write a theorem and its proof as code, and a small **kernel** checks every inference step. If the kernel has no bugs and only accepted axioms are used, passing the check means "this Lean statement is true." The difference from human refereeing is like the difference between a compiler with a type system and code review — the latter misses things; the former doesn't miss *what it checks*.

What "has a Lean proof" guarantees: the trust stack, bottom up

① Lean kernel checks every proof step this repo: sorry = 0, added axioms = 0, native_decide = 0	strong
② Only the 3 standard axioms all 416 Comparator configs allow only propext / Quot.sound / Classical.choice	strong
③ Lean statement = what the paper claims? challenge files written by OpenAI (agents); a human must read the definitions	human
④ Are definitions in deps and patches right? 24 patches (largest ~2 MB) + 11 lana-agents/* dependencies of unknown origin	unaudited
⑤ The ~58% with no formalization no machine check at all; the 3 Hodge-related papers withdrawn on 10-07 were here	none

①② are guaranteed by the machine; ③④⑤ still need humans. Audit data: source grep of `lean/OAI/` (122,458 .lean files, ~26M lines) plus Comparator config counts (2026-10-08, commit fd4aeeb); not compiled.

3.1 Our static audit of the Lean library [P]

CHECK	RESULT	MEANING
sorry (unfinished proof)	0	no "fill in later" steps
axiom declarations	0 (all hits are in comments)	no smuggled-in axioms
native_decide / implemented_by / extern	0	no "trust the compiler" backdoors
decide +kernel	≈1.31M lines, 5,848 files	large finite computations done by the kernel — still trusted, but the proofs are enormous
Comparator configs	416, all allowing only the three standard axioms; only 2 enable the second independent kernel nanoda	see below
lean/patches/	24 patches to third-party dependencies (not the Lean kernel, not Mathlib), largest ~2 MB	patches change definitions and proofs in dependencies ⚠
dependencies	30, of which 11 are <code>lana-agents/*</code> repos of unknown origin	an open point in the trust chain ⚠
formalization.yaml	<code>review: status: unchecked</code>	OpenAI's own label: not reviewed

Scope: lean/OAI/ has 122,458 .lean files, ~26M lines; we only grepped the source and counted configs, without compiling (a full build needs many GB of memory and a Linux sandbox). Independent re-checks: Dave Goldblatt ran the ζ 7/8 proof through Comparator and the nanoda kernel (~63 minutes) [repo]; a third party reviewed the statement of the 9/4 matrix-multiplication theorem [repo].

3.2 Comparator: what it guards against, and what it doesn't

The Lean FRO's Comparator takes a "challenge file" (only the theorem statement, with `sorry` as the proof, importing only trusted libraries such as Mathlib) and a "solution module": the theorem proven in the solution must be **the same type** as the challenge statement, using only allowed axioms. It guards against swapping the statement, changing definitions or adding axioms inside the proof. It does **not** guard against the challenge statement itself failing to say what the paper claims — and the challenge files were written by OpenAI (its agents). That is why the "verification status" of each section below focuses on **reading the statement line by line**.

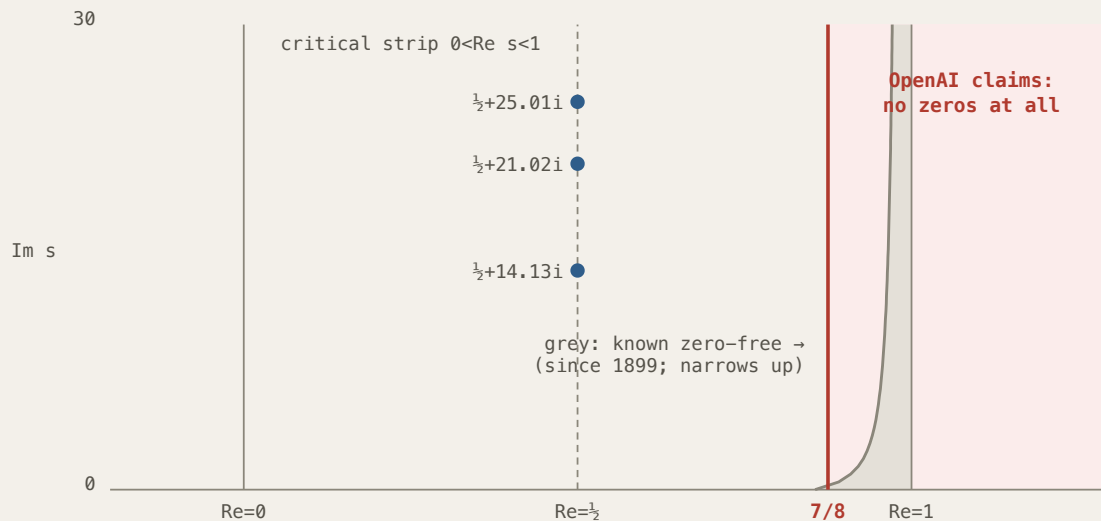
A LIVE COUNTEREXAMPLE

A footnote in OpenAI's plane-coloring paper mentions a May 2026 manuscript that claimed the chromatic number of the plane is 7 with a "formalized theorem" — but that theorem **took a false density bound as a hypothesis**. The formalization was fine; the proven statement was the problem. [P, as reported in the OpenAI paper; original not read ▲] This is the same kind of problem as specification gaming in AI: the system perfectly satisfies the written spec, but the spec is not what you wanted.

NUMBER THEORY · FAMILY 003

4. The quasi-Riemann hypothesis: a ceiling on the noise in the primes

- Level 0** Count primes by hand: 25 below 100, 168 below 1000, 78,498 below 10^6 . Let $\pi(x)$ = "number of primes $\leq x$."
- Level 1** **Prime number theorem:** $\pi(x) \approx \text{Li}(x) = \int_2^x dt/\ln t$. $\text{Li}(10^6) \approx 78,626.5$ vs the true 78,498 — the ratio tends to 1 (proved 1896). The real question is **how big the error $\pi(x) - \text{Li}(x)$ is**.
- Level 2** **The ζ function encodes all the primes:** $\zeta(s) = \sum 1/n^s = \prod_p 1/(1-p^{-s})$. The identity holds because every integer factors uniquely into primes. By hand at $s = 2$: using only $p = 2, 3, 5$ gives 1.5625; all primes below 100 give 1.6419; the true value is $\pi^2/6 \approx 1.6449$.
- Level 3** Extend s to complex numbers $s = \sigma + it$. A **zero** is a point where $\zeta(s) = 0$. The non-trivial zeros all lie in the "critical strip" $0 < \text{Re } s < 1$; the first three are at $\frac{1}{2} + 14.13i$, $\frac{1}{2} + 21.02i$, $\frac{1}{2} + 25.01i$. **Riemann hypothesis (1859):** all of them lie on $\text{Re } s = \frac{1}{2}$.
- Level 4** **Why zeros control the error:** in the explicit formula each zero $\rho = \beta + i\gamma$ contributes an "oscillating noise" term of size about x^β . All zeros with $\text{Re } \rho \leq \theta \Rightarrow$ error about x^θ . The Riemann hypothesis $\Leftrightarrow$ error about $\sqrt{x}$. Think of the zeros as the "spectrum" of the primes, with β setting how fast each frequency's amplitude grows.
- Level 5** **Previously known zero-free regions:** de la Vallée Poussin (1899) $\sigma > 1 - c/\log t$; Vinogradov–Korobov (1958) somewhat wider. **Key point: their width shrinks to 0 as the height $t \rightarrow \infty$.** No one had ever proven that a *fixed* $\delta > 0$ exists with no zeros in the whole strip $\text{Re } s > 1 - \delta$.



Blue dots: the first three non-trivial zeros of ζ (all on $\text{Re } s = \frac{1}{2}$; the Riemann hypothesis says every one is). Grey band: schematic de la Vallée Poussin-type zero-free region (its width shrinks to 0 as the height t grows; curve shape illustrative, constant 0.1). Red region $\text{Re } s > 7/8$: the fixed-width zero-free region claimed by OpenAI family 003. Nobody had previously proven any fixed-width zero-free strip.

4.1 The problem

Quasi-Riemann hypothesis: there is a fixed vertical line $\text{Re } s = \theta < 1$ with no zeros to its right. It is a weak form of the Riemann hypothesis ($\theta = \frac{1}{2}$), but before 2026 it was just as far out of reach.

4.2 Why it was hard

Classical methods start from positivity of the Euler product for $\sigma > 1$ (e.g. the trick $3 + 4\cos \theta + \cos 2\theta \geq 0$), and that positivity weakens with height, so the regions obtained necessarily get narrower. The other route, zero-density estimates (e.g. Guth–Maynard 2024, arXiv:2405.20552), can say "there are few zeros with $\text{Re } s > \sigma$," but not "there are none." A fixed width needs a genuinely new mechanism.

4.3 What OpenAI claims

CLAIM [P]

Main paper (2026-09-30, 199 pages): "We prove that all finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$ and all Dirichlet L-functions are zero-free in the half-plane $\Re s > 7/8$... In particular, the Riemann zeta function is zero-free in this half-plane." The paper stresses: "The Riemann hypothesis remains open."

The family also includes a separate proof of 11/12 (49 pages, written with human assistance) and a 9-page "uniform exclusion of Landau–Siegel zeros."

Method sketch (introduction only): pass to $\mathbb{Q}(\sqrt{-3})$ and build a "completed sum" from Fourier coefficients of the cubic theta series, which can be evaluated two ways (a reflection formula and Poisson summation); a zero with $\beta > 7/8$ would make the two estimates contradict each other. Finally the factorization $L_F(s, \chi \circ N) = L(s, \chi) \cdot L(s, \chi \chi_{-3})$ transfers the result to all Dirichlet L-functions.

4.4 Verification status

Machine

- The Lean challenge statement uses only standard Mathlib definitions, e.g. `theorem riemannZeta_ne_zero_of_seven_eighths_lt_re {s : ℂ} (hs : 7/8 < s.re) : riemannZeta s ≠ 0`. [P]
- Solution ~2,926 files, 487k lines; `sorry/axiom/native_decide` all 0. [P]
- Independent re-check: Goldblatt ran it under two kernels; `#print axioms` lists only the three standard axioms. [P]
- $\triangle$ The dependency patch `PrimeNumberTheoremAnd-lean4341.patch` is ~40k lines and replaces two upstream sorries with proofs — part of the "dependency library" is OpenAI's own code.
- The paper's later applications (Section 6 corollaries) are outside the formalization.

Humans

- Alex Kontorovich (Rutgers; founder of the PrimeNumberTheoremAnd formalization project): "Quasi-RH?!?!?! Are you kidding me? If a human did this, it would be an instant Fields Medal..." [R]
- Levent Alpöge: "Big, big, big, big props for quasiriemann and no Siegel zeroes." [R]
- As of 10-08, **no human expert has publicly said they read all 199 pages.** $\triangle$

Assessment: as long as Mathlib's definition of `riemannZeta` is correct and the Lean kernel has no bug, the theorem is "proven in the machine sense" — an order of magnitude more trustworthy than an unrefereed paper. The remaining risk sits in the build chain (patches and dependencies).

4.5 If true

- **A power saving in the prime number theorem:** $\pi(x) - \text{Li}(x) = O(x^{7/8} \log^2 x)$. Going from "width shrinking to 0" to "fixed width" is a qualitative change; but between $7/8$ and $1/2$ everything is still unknown.
- **Siegel zeros ruled out:** an obstacle that since 1935 made many constants ineffective disappears (e.g. effective lower bounds for class numbers of imaginary quadratic fields).
- **Algorithmic corollaries** (paper Cor. 1.2): least quadratic non-residue $\leq C(\log p)^{32} \Rightarrow$ square roots mod p and the Miller primality test become *unconditional* deterministic polynomial-time algorithms. **No impact on practical cryptography** (randomized algorithms are used in practice, AKS 2002 already gave unconditional deterministic primality testing, and RSA and elliptic-curve security are unaffected).

NUMBER THEORY · FAMILY 017 (+ 005)

5. How well can fractions approximate π ?

Level 0 $22/7$ differs from π by 1.26×10^{-3} ; $355/113$ (Zu Chongzhi, 5th century) by only 2.67×10^{-7} . Compare the benchmark $1/q^2$: $1/113^2 \approx 7.8 \times 10^{-5}$ — $355/113$ beats it by nearly 300×.

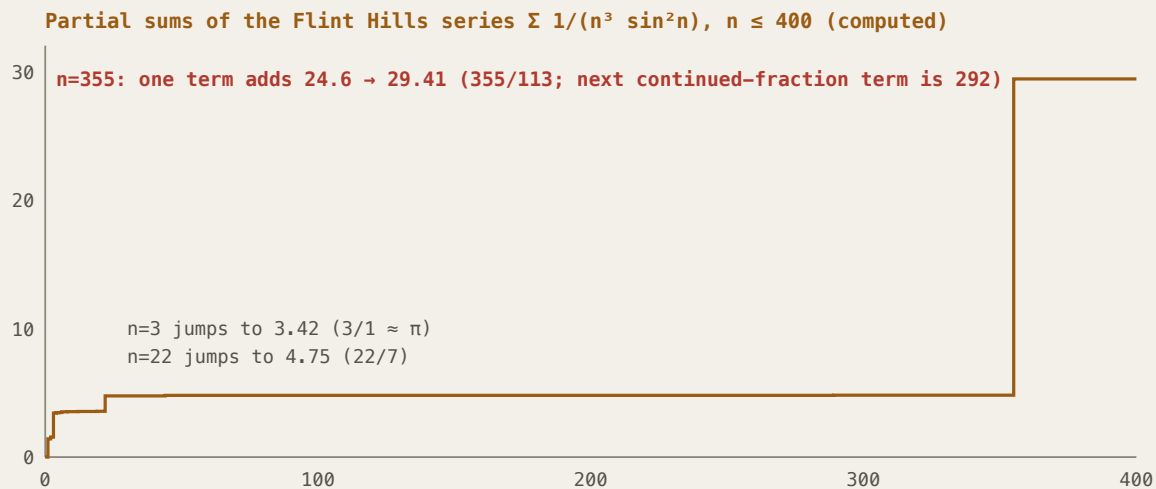
Level 1 **Continued fractions:** $\pi = [3; 7, 15, 1, 292, 1, 1, \dots]$. The larger the next term, the "unusually better" the previous truncation. $355/113$ is followed by 292, which is why it is so accurate.

Level 2 **Dirichlet (1842):** every irrational number has infinitely many p/q with $|x - p/q| < 1/q^2$ (pigeonhole principle). So "exponent 2" is always achievable.

Level 3 **Irrationality exponent** $\mu(x)$ = the largest v such that $|x - p/q| < q^{-(v)}$ has infinitely many solutions. Rationals: 1; Liouville numbers: ∞ ; **algebraic irrationals (e.g. $\sqrt{2}$): exactly 2** — Roth's theorem (1955), Fields Medal 1958; e : 2; almost every real number: 2. The hard part is proving it for *a specific number*.

Level 4

History of upper bounds for π : Mahler 1953: 42 → Mignotte 1974: 21 (20 for large enough denominators) → Hata 1993: 8.016 → Salikhov 2008: 7.606 → Zeilberger–Zudilin 2020: 7.103. Seventy years took it from 42 to 7.1 — still far from 2.



Each jump happens when n is the numerator of an unusually good fraction approximating π (so $\sin n$ is tiny). Whether the series converges depends on how often and how large these jumps are, which is governed by the irrationality exponent of π . Values computed by the build script of this deck.

5.1 What OpenAI claims

CLAIM [P]

The irrationality exponent of π is 2 (2026-09-24, 23 pages): "We prove the conjecture that the irrationality exponent of π is 2. As a consequence, the classical Flint–Hills series $\sum 1/(n^3 \sin^2 n)$ converges."

Precisely: for every $\nu > 2$ there is $Q(\nu)$ such that every $q \geq Q(\nu)$ satisfies $|\pi - p/q| \geq q^{(-\nu)}$. The paper notes that $Q(\nu)$ is *ineffective*, and that $\mu = 2$ is not the same as "bounded partial quotients" (the stronger statement remains open).

Method: suppose there are infinitely many approximations with exponent $\nu > 2$; pick several with well-separated denominators and do multivariate interpolation along a logarithmic curve to get a non-zero determinant. Arithmetically it is a non-zero integer (≥ 1); analytically the approximation errors make it tiny — contradiction. Convergence of Flint Hills then follows from an existing human result (Meiburg 2022: $\mu(\pi) < 5/2 \Rightarrow$ convergence).

The **reasoning summary** (42 pages) is worth reading: the model's initial target was only $\mu < 5/2$; it tried dozens of routes (BBP, modular forms, Padé, p-adic ...), repeatedly hit a wall of

"height vs denominator cost," and finally used weighted interpolation determinants to reach $\mu = 62/25 = 2.48$ first, then 2. It also borrowed determinant ideas from its own Catalan's-constant manuscript.

5.2 Verification status

Lean: `lean/docs/017.md` and `ComparatorChallenges/PiExponent` exist; the solution is ~869 files, 95k lines, grep finds no sorry/axiom [P]; but it is **not listed in formalization.yaml** $\triangle$ and has no third-party re-check. The Flint Hills corollary is outside the official Comparator scope (the solution folder does contain a proved `flint_hills_summable`, but Comparator does not check it); the solution depends only on Mathlib, with no patched libraries. No specific comment from number theorists found $\triangle$.

SIDEBAR: CATALAN'S CONSTANT IS IRRATIONAL (FAMILY 005)

$G = 1 - 1/3^2 + 1/5^2 - 1/7^2 + \dots \approx 0.91597$. In 1978 Apéry proved $\zeta(3)$ irrational (van der Poorten called it "A proof that Euler missed"), but the same idea fails for G : after clearing denominators the linear forms do not tend to 0. The best previous result was "at least one of $\beta(2), \beta(4), \dots, \beta(10)$ is irrational." OpenAI claims G is irrational (44 pages) with Lean (~574k lines, also not in the yaml $\triangle$). If it holds, it is one of the most important single-constant irrationality results since Apéry.

5.3 If true

For the first time a specific constant like π gets the optimal exponent that almost every real number has; the 70-year race of upper bounds ends; a Flint Hills puzzle popular for two decades is settled. Practical applications are minimal — the value is in pure mathematics, and in whether "weighted interpolation determinants" extend to constants such as $\zeta(3)$ and $\log 2$.

THEORETICAL CS · FAMILY 107

6. How fast can matrices be multiplied? $\omega \leq 9/4$

GPU training is, at its core, matrix multiplication. This section asks how fast it can be in principle.

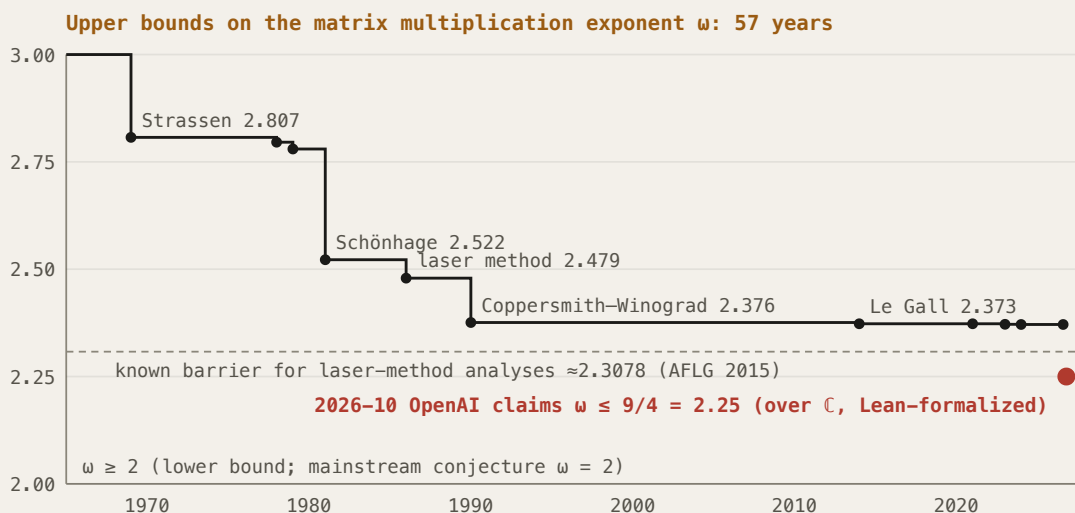
Level 0 Textbook algorithm: $c_{ij} = \sum_k a_{ik} \cdot b_{kj}$ — n^2 entries $\times$ n multiplications each = n^3 . 2×2 needs 8 multiplications.

Level 1 **Strassen 1969: 2×2 with only 7 multiplications** (see the code block below; check it by hand). The algorithm makes no assumption about what $a \dots h$ are, so they can be matrix blocks and the algorithm can recurse: $T(n) = 7T(n/2) \Rightarrow n^{(\log_2 7)} \approx \mathbf{n^{2.807}}$. The one saved multiplication compounds at every level of recursion.

Level 2 **Definition of ω :** the smallest exponent such that $n \times n$ matrix multiplication takes about n^ω operations. Clearly $2 \leq \omega \leq 3$ (lower bound 2: just reading $2n^2$ numbers takes that many steps). The mainstream conjecture is $\omega = 2$.

Level 3 " 2×2 with 7 multiplications" is equivalent to "the 2×2 matrix multiplication **tensor has rank ≤ 7** ." Then: Bini 1979 border rank $\rightarrow$ Schönhage 1981 $\rightarrow$ Strassen 1986's **laser method** (carving many independent matrix multiplications out of a high power of a low-rank tensor). Nearly all progress of the last 40 years came from refining the laser method.

```
A = [[a,b],[c,d]]    B = [[e,f],[g,h]]
M1=(a+d)(e+h)    M2=(c+d)e    M3=a(f-h)    M4=d(g-e)
M5=(a+b)h        M6=(c-a)(e+f)        M7=(b-d)(g+h)
C11=M1+M4-M5+M7    C12=M3+M5    C21=M2+M4    C22=M1-M2+M3+M6
# check C12: M3+M5 = af-ah+ah+bh = af+bh ✓
```



Data: Strassen 1969; Pan/Bini/Schönhage 1978–81; Strassen 1986; Coppersmith–Winograd 1990; Le Gall 2014; Alman–Vassilevska Williams 2021; Duan–Wu–Zhou 2023; Alman et al. 2024 ([arXiv:2404.16349](https://arxiv.org/abs/2404.16349)); Dupont et al. 2026-08 ([arXiv:2608.16884](https://arxiv.org/abs/2608.16884), using AlphaEvolve). From 2014 to 2026 the bound fell by about 0.0017 in total. The claimed 2.25 drops it by about 0.12 at once and lands below the known laser-method barrier, so it must use a different method: the paper uses the asymptotic-spectrum approach of Strassen.

6.1 What OpenAI claims

CLAIM [P]

An Upper Bound of $9/4$ for the Matrix Multiplication Exponent (2026-10-02, **only 13 pages**): "We prove that the exponent of matrix multiplication over the complex numbers is at most $9/4$." The method is not the laser method but Strassen's **asymptotic spectrum** approach: inequalities for "tensor characters" on polynomial-multiplication tensors, ending in $\omega \leq 9/4$. The paper says: "its proof does not specify a competitive finite matrix size."

Two companions in the family: $\omega < 2.258$ over $\mathbb{C}$ (83 pages); $\omega < 2.371054886$ over *every field*, including positive characteristic (35 pages, still in the laser-method framework, about 1.2×10^{-4} better than the previous record).

6.2 Verification status

Statement review [P]: the Lean theorem is `Arithmetic.omega  $\mathbb{C} \leq 9/4$` . Algorithms are modeled as straight-line programs (constant/input/ $+/ - / \times$ gates, each of cost 1), **Correct** requires outputting $A \cdot B$ for *all* input matrices, and ω is the standard non-uniform algebraic-complexity definition — consistent with the literature, no hidden assumptions. The solution directory has no sorry/axiom. **Humans**: Aaronson: "a *rational* exponent for once (!), and via a completely different approach" [R]; no public comment yet from laser-method experts (Vassilevska Williams, Alman, Le Gall) $\triangle$.

6.3 If true – and what it means for AI training

- **In theory**: solving linear systems, matrix inversion, determinants, transitive closure, bipartite matching, linear-programming solvers and more all have complexities written directly in terms of ω , and would all improve. More importantly it shows that the asymptotic-spectrum route works, making $\omega = 2$ look closer.
- **For GPU / LLM training: essentially no direct impact**. This is a classic "galactic algorithm": huge constants, advantageous only on astronomically large matrices. GPU GEMM uses n^3 blocked algorithms, bottlenecked on memory bandwidth and Tensor Core throughput; even Strassen is rarely used in practice (poor numerical stability, irregular memory access).
- **How this differs from AlphaTensor / AlphaEvolve**: those find low-rank decompositions of *fixed small sizes* (e.g. AlphaEvolve's 2025 4×4 complex scheme with 48 multiplications) — constant-factor optimizations that can land in real kernels. ω is about the *exponent* as $n \rightarrow$

∞ . Dupont et al. (2026-08) used AlphaEvolve to tune laser-method parameters, the first time AI touched ω , for a gain of 1.6×10^{-4} .

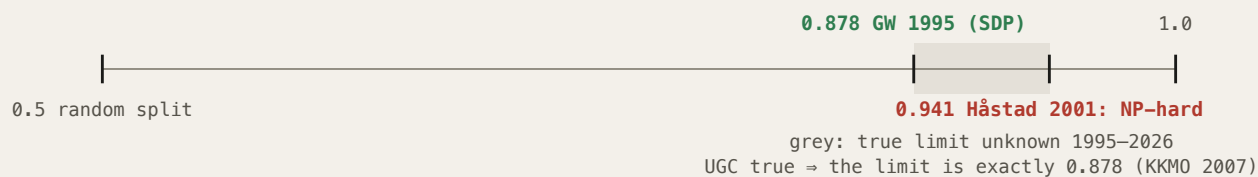
THEORETICAL CS · FAMILY 102

7. Unique Games: where is the ceiling for approximation algorithms?

- Level 0** **NP-hard:** problems like 3SAT and Max-Cut are believed to have no exact polynomial-time algorithms ($P \neq NP$).
-
- Level 1** **Settle for approximation.** Max-Cut: split a graph's vertices into two groups to maximize edges crossing between them. A random split cuts half the edges in expectation. By hand on a triangle: the best cut has 2 edges, a random split expects 1.5, ratio 0.75.
-
- Level 2** **Goemans–Williamson (1995):** relax each vertex to a unit vector, solve a semidefinite program (SDP), then cut with a random hyperplane — guaranteeing **0.878**.
-
- Level 3** **Approximation can be hard too:** the PCP theorem (1992–98). Håstad (2001) showed approximating Max-Cut beyond $16/17 \approx 0.941$ is NP-hard. That left a 25-year gap between 0.878 and 0.941.
-
- Level 4** **Unique Games:** each vertex picks one of q labels; each edge is a permutation constraint (fix one end and the other is determined). By hand: a triangle, labels $\{0,1,2\}$, constraints $b=a+1$, $c=b+1$, $a=c+1 \pmod{3}$ — going around gives $a=a+3=a$, fully satisfiable; change the last to $a=c+2$ and at most $2/3$ can be satisfied. Fully satisfiable instances are solvable in polynomial time by propagating along edges; **only "almost satisfiable" is hard**.

UGC (Khot 2002): for every ϵ , $\delta > 0$ **there is** an alphabet size k (chosen after ϵ and δ) such that distinguishing " $\geq 1-\epsilon$ satisfiable" from "at most δ satisfiable" is NP-hard.

Max-Cut approximation ratio: what fraction of the optimum can polynomial time guarantee?



Unique Games completeness ladder: which instances are provably NP-hard?

≈1/2 completeness
2-to-2 theorem, 2018



1-ε completeness = UGC
OpenAI 2026-09 claim + Lean

Fully satisfiable instances
are easy (propagate along
edges): only "almost" is hard

Sources: Goemans–Williamson, JACM 1995; Håstad, JACM 2001; Khot, STOC 2002; Khot–Kindler–Mossel–O'Donnell, SICOMP 2007; Khot–Minzer–Safra, FOCS 2018.

7.1 Why it matters so much

- **KKMO 2007**: if UGC holds, the limit for Max-Cut is *exactly* 0.878 — GW is optimal.
- **Khot–Regev 2008**: if UGC holds, approximating Vertex Cover to $2-\epsilon$ is hard.
- **Raghavendra 2008**: if UGC holds, for *every* constraint satisfaction problem the basic SDP is the optimal polynomial-time approximation algorithm. One algorithm plus one conjecture fix the approximation limit of a whole class of problems.
- Evidence against: Arora–Barak–Steurer 2010 gave a subexponential-time algorithm for UG, which made some doubt UGC. The most recent progress was the 2018 **2-to-2 theorem** (Khot–Minzer–Safra et al.), proving the version with completeness about 1/2 — "half of UGC."

7.2 What OpenAI claims

CLAIM [P]

The Unique Games Theorem (2026-09-23, 58 pages): "We prove the Unique Games Conjecture. For every fixed $\epsilon, \delta \in (0, 1/2)$, we give a deterministic polynomial-time reduction from 3SAT to Unique Games over a fixed finite alphabet, with completeness at least $1 - \epsilon$ and soundness at most δ ."

The family also contains four direct reductions that *do not depend on UGC*: Max-Cut beyond 0.878 is NP-hard, Vertex Cover below 2 is NP-hard, and so on — which is what Aaronson meant when he observed (paraphrased) that UGC was not actually needed for its most famous consequences.

7.3 Verification status

Statement review [P]: the Lean theorem is `OAI.UniqueGamesTheorem.theorem11`. The input is a binary encoding of a 3SAT formula; the reduction must satisfy `Turing.TM2ComputableInPolyTime` (Mathlib's multi-tape Turing machines); completeness and soundness are stated cleanly, and the instances are additionally required to be "translation constraints on simple bipartite graphs" — **a stronger, known-equivalent form of UGC; the encoding is faithful**. Solution ~478 files, 187k lines, sorry/axiom 0.

Humans [R]: Lance Fortnow: "if they hold up, we've seen more progress in TCS in the last 24 hours than in the previous three decades combined." Mark Braverman (Quanta): "Math by press release is not that healthy for math." Dana Moshkovitz reportedly criticized the paper as extremely hard to read without AI help (second-hand, and sources word it differently [△](#)). No public statement from Khot himself [△](#).

7.4 If true

For a large class of problems, the question "how well can efficient approximation do?" gets an exact boundary; hundreds of "if UGC then ..." papers become theorems. One association for AI safety (*my speculation, not an established conclusion*): the complexity-theoretic basis of debate-style scalable oversight (Irving–Christiano–Amodei 2018) is interactive proofs and PCPs; UGC pins down the boundary of "a little error makes efficient distinction impossible," which is suggestive for protocols where a weak verifier oversees a strong prover.

THEORETICAL CS · FAMILY 103 (+ 109)

8. $L = RL = BPL$: in small memory, randomness doesn't help

Level 0 **What does randomness buy?** To test whether $(x+y)^2 = x^2+2xy+y^2$ holds identically, plug in a few random numbers — very fast. A deterministic fast algorithm is still unknown.

Level 1 **P vs BPP:** the mainstream conjecture is $P = BPP$ (randomness gives no essential speed-up), but an unconditional proof is far away.

Level 2 **The memory-bounded version:** L = deterministic computation with only $O(\log n)$ bits of memory (roughly a few pointers); RL/BPL = the same memory plus random bits. By

hand: a random walk of n^3 steps from s in an undirected graph needs only "current vertex + counter" and reaches t with high probability $\Rightarrow$ connectivity $\in$ RL (1979).

Level 3

History of derandomization: Borodin–Cook–Pippenger 1983 $BPL \subseteq L^2$ (Nisan 1992 recovered it with an $O(\log^2 n)$ -seed generator, and in 1994 got polynomial time in the same space) $\rightarrow$ Saks–Zhou 1999 $BPL \subseteq L^{3/2}$ $\rightarrow$ Hoza 2021 slightly better $\rightarrow$

Reingold 2005: undirected connectivity $\in$ L (a milestone in removing randomness from a specific problem). General BPL stayed near the $3/2$ power for 25 years.

CLAIM [P]

Exact Derandomization of Logarithmic Space: $L = RL = BPL$ (2026-09-23, 108 pages):

"We prove $L = RL = BPL$, resolving the derandomization problem for polynomial-time randomized logarithmic space." A stronger version: acceptance probabilities can be estimated deterministically to precision 2^{-q} in space $O(\log n + q)$.

Verification [P]: the Lean theorem is $L = RL \wedge RL = BPL$; for this family Comparator additionally **locks 20 definitions** (machine, step, space, acceptance probability, L/RL/BPL), so the solution cannot quietly change them. The model is a custom multi-tape Turing machine, space counts work cells visited, and "every coin sequence halts in polynomial time" — the standard BPL definition. sorry/axiom 0. **Humans** [R]: Aaronson called it "one of the great derandomization conjectures short of $P=BPP$," adding "its truth was never in serious doubt" — nobody doubted the statement; the difficulty was the proof.

If true: the first natural resource class with randomness removed completely and unconditionally. It does **not** imply $P = BPP$ and is far from P vs NP (Fortnow: "none of these results get us any closer to settling $P \neq NP$ "). The proof's "low-space, high-precision linear algebra" may matter more than the result.

SIDEBAR: INTEGER MULTIPLICATION BELOW $N \log N$ (FAMILY 109)

Schoolbook n^2 ; Karatsuba 1962 $n^{1.585}$; Schönhage–Strassen 1971 $n \log n \log \log n$, conjecturing $n \log n$ optimal; Harvey–van der Hoeven 2019 (Annals 2021) reached $O(n \log n)$. OpenAI claims $O(n (\log n)^{(1-2^{-182})})$ on multi-tape Turing machines (73 pages, **no Lean**). A saving of 2^{-182} is visible only at astronomical sizes — no practical significance — but it refutes a 50-year "optimality" conjecture (in that model of computation).

9. Coloring the plane: points at distance exactly 1 must differ

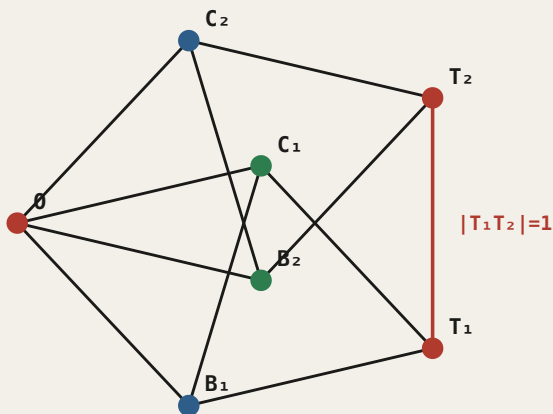
Level 0 **Graph coloring** (familiar from register allocation and scheduling): adjacent vertices get different colors; how many colors are needed?

Level 1 **The unit-distance graph:** *every point* of the plane is a vertex; points at distance exactly 1 are joined. Its chromatic number is written $\chi(\mathbb{R}^2)$.

Level 2 **Why 3 colors are not enough:** the Moser spindle (figure below, 7 points). **Why 7 are enough:** tile the plane with hexagons of diameter slightly below 1 and color them periodically with 7 colors; same-colored hexagons are more than 1 apart.

Level 3 So $4 \leq \chi \leq 7$, **stuck from the 1950s until 2018**, when the biogerontologist Aubrey de Grey pushed the lower bound to 5 with a 1581-vertex finite graph (in Polymath16, Heule used a SAT solver with unsat-core extraction to get 553 vertices, and Parts then minimized the graph to 509).

Moser spindle (1961): 7 points, 11 unit-length edges $\Rightarrow$ 3 colors are not enough



Argument (with 3 colors)

O, B_1, C_1 pairwise at distance 1 $\rightarrow$ one color each
 T_1 is 1 from $B_1, C_1 \rightarrow T_1 = \text{color of } O$
 likewise $T_2 = \text{color of } O$
 but $|T_1 T_2| = 1 \rightarrow \text{contradiction} \Rightarrow \chi \geq 4$

Known bounds on χ of the plane

1950s $4 \leq \chi \leq 7$ (spindle, hexagons)

2018 $5 \leq \chi \leq 7$ (de Grey graph)

2026-09 $6 \leq \chi \leq 7$ (OpenAI, Lean)

Coordinates computed by the build script: two rhombi, each made of two unit equilateral triangles, share the vertex O , and each tip T is $\sqrt{3}$ from O . Rotating one rhombus about O by $2 \cdot \arcsin(1/(2\sqrt{3})) \approx 33.6^\circ$ puts the two tips exactly 1 apart.

CLAIM [P]

The Euclidean plane is not five-colorable (2026-09-23, 62 pages): "We prove that every coloring of the Euclidean plane with five colors has a monochromatic unit-distance pair, with no regularity assumption on the color classes. Consequently, the chromatic number of the plane is either six or seven."

A surprising method: the standard lower-bound recipe is "find a finite graph and prove it is not k -colorable" (de Grey's route). This paper **exhibits no finite graph at all**, and uses ergodic theory and measure theory instead: first it proves "a genuine k -coloring exists $\Leftrightarrow$ a 'weakly measurable' k -coloring exists" (by averaging over translations and rotations, using the *amenability* of those groups — which connects to §10), then shows no weakly measurable 5-coloring exists. The last step returns to the 1961 Moser spindle.

Verification [P]: the Lean statement is two lines a person can check at a glance:

```
def ProperColoring (k) (c :  $\mathbb{C} \rightarrow \text{Fin } k$ ) : Prop :=  $\forall x\ y : \mathbb{C}, \|x - y\| = 1 \rightarrow c\ x \neq c\ y$   
theorem no_proper_five_coloring :  $\neg \exists c : \mathbb{C} \rightarrow \text{Fin } 5, \text{ProperColoring } 5\ c$ 
```

No measurability assumption — **faithful to the original problem**. The proof is ~70 files, 30k lines, no sorry/axiom. Gil Kalai: "raises the lower bound to six, leaving six and seven as the two possible answers," while cautioning that "even Lean verification may have issues" [R].

If true: $\chi(\mathbb{R}^2) \in \{6, 7\}$. A new open problem: by the de Bruijn–Erdős theorem a finite unit-distance graph needing 6 colors must exist, but nobody knows what it looks like — a good target for AI plus SAT. It is also the ideal division of labor: humans review a two-line statement, the machine checks 30,000 lines of proof.

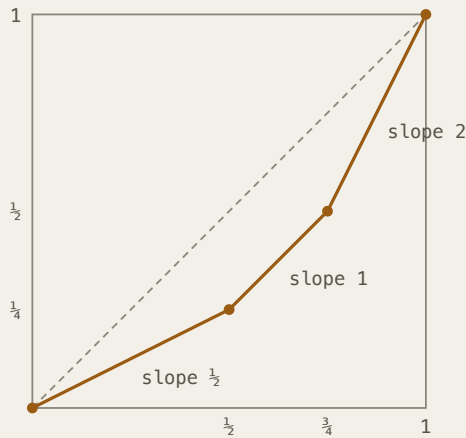
SIDEBAR: BORSUK'S CONJECTURE FAILS IN DIMENSION 9 (FAMILY 156)

A disk cannot be cut into 2 pieces of smaller diameter (one piece always contains an antipodal pair), but 3 pieces suffice. Borsuk asked in 1933: can every bounded set in $\mathbb{R}^d$ be cut into $d+1$ pieces of smaller diameter? Kahn–Kalai disproved it in 1993 in dimension 1325; the dimension of counterexamples later fell to 64 (2014) and 63 (2026-05, with help from GPT-5.5 Pro). OpenAI claims that the projection matrices uu^T of all unit vectors u in $\mathbb{R}^4$, which live in a 9-dimensional space, **cannot be covered by 10 pieces of smaller diameter** — from 63 straight down to 9 (18 pages, with Lean). Dimensions 4–8 remain open.

10. Thompson's group F : a problem "proven" and withdrawn for decades

- Level 0** A **group** is a set of operations that can be composed and undone. Examples: integer addition; the 8 symmetries of a square.
-
- Level 1** **Generators:** a few operations build the whole group. $\mathbb{Z}$ is generated by "+1"; the **free group** F_2 is generated by a and b with no relations at all, and its Cayley graph is an infinite tree with 4 branches at every vertex.
-
- Level 2** **Thompson's group F :** increasing piecewise-linear maps of $[0,1]$ to itself, with breakpoints at $k/2^n$ and slopes powers of 2; the operation is composition. Generated by x_0 (figure below) and x_1 .
-
- Level 3** **Amenable = can you "average fairly"?** Følner criterion: are there finite sets A whose proportion of change under translation by the generators is arbitrarily small? $\mathbb{Z}$: yes; F_2 : no. Von Neumann introduced the notion in 1929 to explain the **Banach–Tarski paradox** (a ball cut into finitely many pieces reassembled into two balls of the same size) — the paradox comes from a copy of F_2 inside the rotation group.
-
- Level 4** "Is every non-amenable group non-amenable because it contains F_2 ?" No (counterexamples since Ol'shanskii 1980). The early ones were exotic; since 2013 Monod and Lodha–Moore have given natural, finitely presented, torsion-free counterexamples. F contains no F_2 (Brin–Squier 1985), so whether F is non-amenable remained the most famous test case on this line.

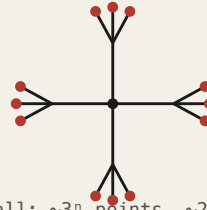
Left: the generator x_0 of the Thompson group F · Right: can you average fairly?



$\mathbb{Z}$: amenable

$A = \{-n, \dots, n\}$: shift by 1 changes $2, 2/(2n+1) \rightarrow 0$

Free group F_2 : non-amenable



radius- n ball: $\sim 3^n$ points, $\sim 2/3$ on the boundary

x_0 squeezes $[0, 1/2]$ onto $[0, 1/4]$, shifts $[1/2, 3/4]$ onto $[1/4, 1/2]$ and stretches $[3/4, 1]$ onto $[1/2, 1]$; F is generated by x_0 and x_1 . F contains no free subgroup F_2 (Brin–Squier 1985), so the standard key "contains $F_2 \Rightarrow$ non-amenable" does not apply. That is one reason the problem stayed open for 47 years.

10.1 The best case for why Lean matters

Geoghegan conjectured in 1979 that F is non-amenable. Since then proofs have been announced in both directions: in 2009 two arXiv papers appeared almost simultaneously, Akhmedov (non-amenable) and Shavgulidze (amenable); in 2011 Moore showed the latter's errors were "serious and do not seem to be repairable" — after it had been published;

Justin Moore himself claimed amenability in 2012, then withdrew after an error in

Lemma 4.13 was pointed out; Akhmedov's version has never been accepted. [P, arXiv

1102.0747 / 1209.2063 / 1310.4395]

CLAIM [P]

Thompson's group F is nonamenable (2026-09-23, 13 pages in total including the appendix; the proof itself is about 7): "We prove that Thompson's group F is nonamenable. This confirms Geoghegan's conjecture."

Idea: borrow a phenomenon from functional analysis — the unit ball of an infinite-dimensional Hilbert space carries a Lipschitz map that moves every point by at least $1/2$ (Benyamini–Sternfeld 1983). If F had Følner sets, averaging over them would yield an approximate fixed point of that map — contradiction. The key tool comes from infinite-dimensional geometry, not the combinatorial counting usual in group theory.

Verification [P]: the Lean challenge file **defines F from scratch** (increasing homeomorphisms, finitely many dyadic breakpoints, slopes 2^k) and an "invariant mean"

(linear, positive, normalized, left-invariant); the theorem says "there is a group structure whose multiplication is composition, and no invariant mean exists." Multiplication is forced to be composition, leaving no room to cheat — **faithful**. Proof ~68 files, 9,500 lines, no sorry/axiom. Kalai cautiously says the paper "announces" the result [R]; no public comment yet from Geoghegan, Moore or Akhmedov $\triangle$.

Takeaway: the human history of this problem is a chain of "announce $\rightarrow$ find error $\rightarrow$ withdraw," including from top experts. The statement file is 89 lines including imports, auditable line by line; for a 13-page paper, past experience suggests the community might need years to reach consensus. Lean replaces "trust the author" with "trust the kernel and the statement."

PDE · SEPARATE SEPTEMBER RELEASE (+ FAMILY 376)

11. Navier–Stokes: can a fluid "blow up" in finite time?

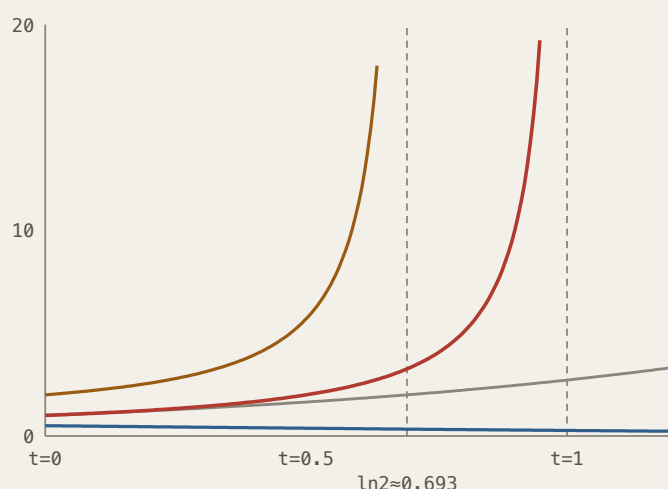
Level 0 **ODE vs PDE:** an ODE's unknown is a single number changing in time, $x(t)$; a PDE's unknown is a whole field, such as the velocity $u(x,t)$ at every point and time — like infinitely many coupled ODEs.

Level 1 **Blowup by hand** (figure below): $\dot{x} = x^2$, $x(0) = 1$ has solution $1/(1-t)$: 10 at $t = 0.9$, 100 at $t = 0.99$, ∞ as $t \rightarrow 1$. The equation is perfectly smooth, yet the solution breaks in finite time. Add damping, $\dot{x} = x^2 - x$: small data decays, large data still blows up.

Level 2 **The NS equations:** $\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f$, $\nabla \cdot u = 0$. $(u \cdot \nabla)u$ is nonlinear self-transport (the x^2 of the toy), $\nu \Delta u$ is viscosity (the $-x$), f is external forcing.

Level 3 **Why 3D is hard:** the only global conserved quantity is energy, and under 3D scaling the energy bound controls smaller scales less and less — "supercritical." 2D is critical, which is why 2D was settled long ago.

Finite-time blowup by hand: a smooth equation whose solution reaches infinity in finite time



- $\dot{x} = x: e^t$, grows but never blows up
- $\dot{x} = x^2: 1/(1-t)$, blows up at $t=1$
- $\dot{x} = x^2 - x, x_0=2$: blows up at $t=\ln 2$
- $\dot{x} = x^2 - x, x_0=\frac{1}{2}$: decays to 0

Clay alternatives (any one wins)

- (A) $\mathbb{R}^3$, no force: smooth forever
- (B) $\mathbb{T}^3$, no force: smooth forever
- (C) $\mathbb{R}^3$, chosen smooth force: blowup
- (D) $\mathbb{T}^3$, chosen smooth force: blowup

OpenAI 9/8: (C)/(D); (A)/(B) open

Left: the x^2 term plays the role of the nonlinear self-transport in Navier–Stokes, the $-x$ term the role of viscous damping. With small initial data damping wins (blue); with large data the nonlinearity wins (orange). This mirrors what is known for 3D NS: global smooth solutions for small data, unknown for large data. Curves computed from the exact solutions by the build script. Right: the four alternatives in the official Clay problem statement (Fefferman).

CLAIM [P]

"Finite time blowup for Navier–Stokes" (2026-09-08, 166 pages, plus a separate ~640k-line Lean repository [openai/NavierStokesAndEuler](#)): "For every positive viscosity, we construct a solution ... that starts from rest and develops unbounded velocity in finite time while maintaining uniformly bounded kinetic energy." The forcing f is smooth and compactly supported in space-time; the paper says this establishes Clay's **alternative (C) and implies (D)**.

11.1 The key detail: you choose the force

Section 2 of the paper says it plainly: "For any incompressible flow u and pressure p , we can always define the external force f to be the residual ... The challenge is to choose a flow that blows up while this residual remains smooth." So (C)/(D) turns the problem into "can you build a flow that blows up while the residual force stays smooth?" — much weaker than whether the fluid blows up *on its own* (the unforced (A)/(B)). Most analysts regard (A)/(B) as the real question [R]. This is not cheating: Fefferman deliberately included (C) as "retaining the heart of the problem". The gap between the written problem and what people care about only appeared after 2023, when the forced route turned out to be easier.

11.2 Verification and disputes

- **Lean** [P]: ~2,659 files, 640k lines, no sorry/axiom; Comparator enables the second kernel nanoda. The statement is adapted from DeepMind's Formal Conjectures repo and matches Clay's conditions (4)–(11) clause by clause, including the erratum on pressure in the periodic case. The same repo also formalizes blowup for **unforced** 3D Euler from smooth, compactly supported data — Fefferman calls the Euler problem "also open and very important", so this part may matter as much as NS.
- **Community** [R]: Clay president Bridson said evaluation will be "deliberately unhurried"; Clay still lists NS as unsolved; no journal submission reported. Clay's rules require publication in a qualifying venue plus a two-year acceptance period.
- **Credit dispute** [R]: Buckmaster and Alpöge (NYU) had forced blowup for Euler/Boussinesq in August and posted it on September 7–8; Buckmaster says OpenAI started on this direction after hearing of their progress, while OpenAI denies his prompts influenced its system (accounts differ $\triangle$). Fefferman (author of the Clay statement): "The heroes of the story ... are Córdoba and Martínez-Zoroa" (prior work on forced Euler blowup). The paper (CDN PDF, 2026-09-08) already cites Córdoba–Martínez-Zoroa, but not Buckmaster–Alpöge. Gómez-Serrano (NPR): "The paper is not written for humans ... as of today, the paper doesn't teach us much."

RELATED: UNIVERSAL COMPUTATION IN FORCED NAVIER–STOKES (FAMILY 376)

Claim: on the 3-torus there is a smooth forcing "program" such that a fluid particle enters a target region **if and only if** a given Turing machine halts — so this reachability question is undecidable. The paper is candid: "There is an elementary way to force a chosen smooth incompressible velocity..." It recalls Tao's 2016 program (get the fluid itself to build a computer that hands its energy to ever-smaller copies of itself, causing blowup), but 376 puts the "programming" into the forcing and **does not advance** that unforced-blowup program; it does not cite Tao 2016.

If true: one alternative of a Millennium Prize problem is formally closed. Almost no direct impact on weather, turbulence or engineering — engineering solvers never relied on global regularity, and real fluids hand over to molecular scales long before any singularity.

a plasma model break down by itself?

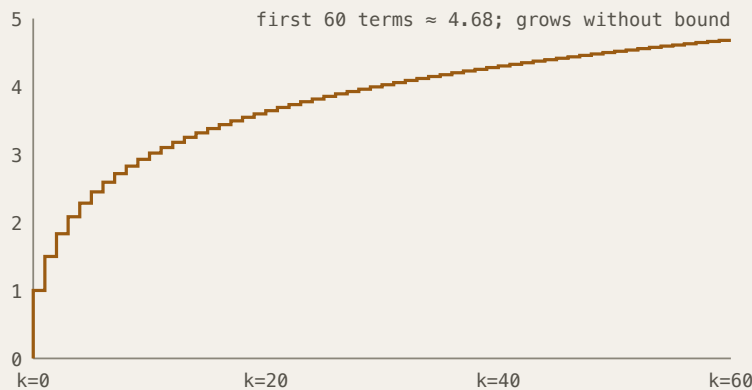
Level 0 **Plasma:** ionized gas (fusion reactors, the solar wind). Too thin for collisions to matter, so it is described by a **phase-space density** $f(t, x, v)$: how many particles are near position x with momentum v — 6 dimensions plus time.

Level 1 **Self-consistent coupling:** particles feel the Lorentz force (Vlasov equation) → particles form currents → currents generate fields (Maxwell equations) → fields push particles. The nonlinearity comes from this loop.

Level 2 **Relativity** ($c = 1$): velocity $\hat{v} = p/\sqrt{1+p^2}$. $p = 1$ gives 0.707, $p = 10$ gives 0.995 — momentum can be arbitrarily large, speed is always < 1 .

Level 3 **Glassey–Strauss 1986:** singularities can only appear at high speed — as long as particle momenta stay bounded on finite time intervals, the solution continues. The whole problem reduces to: **can particles be accelerated to infinite momentum in finite time?** The 40 years since produced only results with restrictions (small data, lower dimension, symmetry).

Vlasov–Maxwell endgame: each momentum doubling takes $\geq 1/k$ time, and $\sum 1/k$ diverges



k = number of momentum doublings so far
doubling k ($P \approx 2^k$) takes $\geq 1/(k \cdot \log 2)$
total time $\approx$ harmonic series $\rightarrow \infty$
 $\Rightarrow$ finitely many doublings in finite time
 $\Rightarrow$ momentum stays bounded, so by
Glassey–Strauss 1986: no blowup

A hand-computable version of the argument skeleton in the reasoning summary of the OpenAI paper (family 362). The real difficulty is proving that each doubling takes at least $\sim 1/\log P$ time (signed momentum-increment estimates plus counting direction changes). Staircase computed by the build script.

CLAIM [P]

"Global classical solutions of the three-dimensional relativistic Vlasov–Maxwell system" (2026-09-23, 49 pages): "We prove global existence and uniqueness for arbitrary smooth admissible initial data in the three-dimensional, one-species relativistic Vlasov–Maxwell system ... without size or symmetry restrictions on the data."

Verification [P]: the hypotheses of the Lean statement `global_classical_solution` were checked one by one (f_0 smooth, compactly supported, non-negative; fields with bounded derivatives in L^2 ; the two Gauss constraints), and the conclusion is existence + smoothness + uniqueness. ~207 files, 90k lines, no sorry/axiom. **Scope limit:** only **one particle species**, while real fusion plasmas have at least two (electrons and ions); whether the extension is routine is unclear $\triangle$. No named comment from plasma-PDE experts found $\triangle$.

The reasoning summary is candid: the model hit walls repeatedly ("exponent inequality contradiction: $H < 0.41z$ vs $H > 0.496z$ "), and its last sentence says the conclusion "rests on the occupation bound, direction-count estimate, and cutoff cancellations ... the cited continuation theorem alone does not establish those estimates."

If true: a flagship problem open for about 40 years (one species) is solved with machine verification — possibly weightier than most results in the batch. No short-term effect on fusion engineering (PIC and gyrokinetic simulations do not depend on this theorem); what it proves is that no particle's momentum in the model goes to infinity in finite time (total energy is conserved anyway; the point is individual particles) — a "model health certificate."

PROBABILITY AND STATISTICAL MECHANICS · FAMILY 221

13. Sparse spin glasses: from magnets to the energy landscapes of neural networks

This section connects directly to neuroscience and machine learning: the energy function of a Hopfield network is the Hamiltonian of a spin glass.

Level 0 **Ising magnet:** a spin $\sigma_i = \pm 1$ at each site, energy $H = -\sum J_{ij} \sigma_i \sigma_j$. With $J > 0$ neighbors want to align.

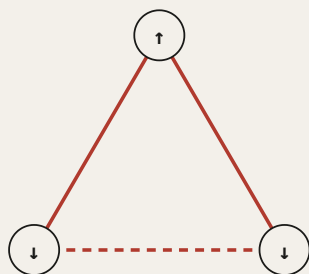
Level 1 **Frustration** (figure, left): three spins pairwise "want" to be opposite; none of the 8 configurations satisfies all three bonds; the ground state is 6-fold degenerate.

Level 2 **Spin glass:** couplings J_{ij} are *random* $\pm$, frustration is everywhere, the energy landscape is rugged with exponentially many metastable states. The **SK model** (1975) couples every pair; **Parisi (1979)** gave its free energy via "replica symmetry breaking" — magical physics, not rigorous math; Guerra 2003 + Talagrand 2006 proved it rigorously; Parisi won the 2021 Nobel Prize in Physics.

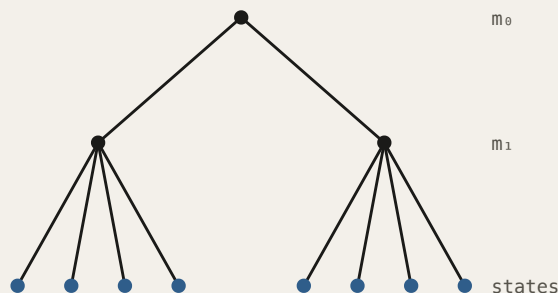
Level 3 **The diluted (sparse) version:** each spin has only finitely many bonds. **Mézard–Parisi (2001)** gave a one-level (1RSB) formula via the cavity method on the fixed-degree Bethe lattice; the general hierarchical form and its upper bound come from Panchenko–Talagrand 2004, but **equality was an open conjecture**. The difficulty: in dense models pairwise overlaps describe the state; sparse models need control of all multi-replica overlaps.

Level 4 **Link to random k-SAT:** k-SAT is a sparse spin glass with k-body interactions. Ding–Sly–Sun (2015) rigorously proved the satisfiability threshold for large k — exactly what these physics methods predicted.

Left: the smallest frustrated system · Right: the hierarchical (ultrametric) structure of Parisi



all three bonds "want" opposite spins
none of the 8 configurations satisfies all three
(dashed bond always fails): 6-fold degenerate
 $Z(\beta) = 2e^{(-3\beta)} + 6e^{\beta}$



closer relatives $\Rightarrow$ larger overlap
diluted: need all multi-replica overlaps

Spin glass = random $\pm$ couplings + frustration everywhere + exponentially many metastable states. The energy function of a Hopfield network (1982) is exactly such a Hamiltonian; perceptron capacity (Gardner 1988) was computed with the same replica method.

CLAIM [P]

"The Mézard–Parisi formula for diluted spin glasses" (2026-09-23, 36 pages): "We prove the Mézard–Parisi hierarchical cavity formula for diluted even-arity Ising models in the Panchenko–Talagrand class." Key trick: shift every branching depth down one level and then average over depths to control multi-replica overlaps. The reasoning summary contains the line: "Aha recursive structure: overfitting shared variable can simulate original model at smaller scale."

Verification [P]: Lean theorem `mezard_parisi` (pressure converges to the variational value), with `Admissible` spelling out the Panchenko–Talagrand hypotheses; ~405 files, 56k lines, no sorry/axiom. **Narrower than the title suggests**: only **even-arity** models with a positivity condition — odd-arity models such as 3-SAT and hard-constraint SAT at zero temperature are excluded, so it does **not** directly extend Ding–Sly–Sun. It also covers only **Poisson-degree** diluted graphs; fixed-degree random regular graphs (the Bethe lattice of the original Mézard–Parisi paper) are not covered. Hopfield networks (fully connected, with pattern-correlated couplings) are outside the theorem too; the connections below are about where the ideas came from, not consequences. No named expert comment found $\triangle$.

13.1 Connections to ML and neuroscience (the honest version)

- **Hopfield networks** (1982): memories are energy minima; storage capacity was computed with the replica method (Amit–Gutfreund–Sompolinsky 1985); Hopfield and Hinton shared the 2024 Nobel Prize in Physics.
- **Perceptrons**: Gardner (1988) computed capacity with replicas; family 222 in the same release gives the Ising perceptron free energy and a jamming exponent.
- **Loss landscapes**: the intuition "rugged landscape with hierarchically clustered minima" comes from spin glasses (e.g. Choromanska et al. 2015). But these analogies are mostly *heuristic*; the theorem does not tell you what SGD will find.

CAUTIONARY CASE · FAMILY 032 AND THE OCT 7 WITHDRAWALS

14. One sign error, three papers withdrawn: the risk of no formalization

The Hodge conjecture (a Millennium Prize problem), minimally: on a smooth projective complex algebraic variety some "topological holes" can be "filled" by subvarieties defined

by algebraic equations; the conjecture says every hole satisfying a certain linear-algebra condition can be filled this way. The case $p = 1$ has been known since 1924 (Lefschetz).

OpenAI claims the rational Hodge conjecture for all **complex-multiplication (CM) abelian varieties** (53 pages) — a special class, not the Millennium problem itself; like the ζ zero-free region it was **not** produced by the fixed procedure, and it has **no Lean**. [P]

Oct 7 withdrawals: one sign error propagates through the dependency graph

Weil classes on abelian eightfolds
sign error: -1 written as $+1$

Kuga-Satake for K3 surfaces
depends on the above → withdrawn

Rational Hodge for products of K3s
depends on the above → withdrawn

Rational Hodge, CM abelian varieties
not on this chain; not withdrawn; no Lean

Family 032 first titled "Hodge and Kuga-Satake ... all K3 surfaces"
→ retitled to the CM-only version afterwards

722 → 719; 14 repaired, 13 re-cited

Sources: `openai/math history.md` (2026-10-07); the family title is compared against `CONTENTS.md` at the initial commit `adc7f12`. The withdrawal notice reads: "This withdrawal concerns the proof; it does not assert that the mathematical statement is false."

14.1 What happened [P]

history.md (10-07): "In 'Algebraicity of Weil classes on split abelian eightfolds' a sign error invalidates a stabilization-trace cancellation argument and the construction used by two dependent papers." A sign that should have been -1 was taken as $+1$; a signed count went from the claimed 0 to $-2m \neq 0$, so the hypothesis of a key theorem failed, and two dependent K3-surface papers were withdrawn with it. The CM paper does not cite the withdrawn paper and is not on this chain.

14.2 Three lessons

1. **Unformalized $\approx$ unverified**: only $\sim 42\%$ of top-line results have Lean. This error was a single sign — missed by human eyes and the model's self-checks, and Lean would certainly have caught it.
2. **Errors propagate through the dependency graph**: one lemma fails, three papers fall, 13 downstream papers need new citations — the same structure as error compounding in long agent reasoning chains.

3. **Title drift:** family 032 was first titled "Hodge and Kuga–Satake results for all projective K3 surfaces" and only downgraded to the CM version after the withdrawals. When reading bulk AI output, trust the latest commit and history.md, not launch-day news.

SYNTHESIS · CROSS-CUTTING VIEW

15. The 10 results side by side

RESULT	PREVIOUS STATE	CLAIMED JUMP	STATEMENT FAITHFUL?	CONFIRMED BY HUMAN EXPERTS
Quasi-RH 7/8	no fixed-width zero-free strip at all	qualitative	yes (standard Mathlib definitions)	none
$\mu(\pi) = 2$	7.103	straight to the end	yes (not in yam1 $\triangle$)	none
$\omega \leq 9/4$	2.371177	−0.12, past a known barrier	yes	none
UGC	"half" (2-to-2, 2018)	complete	yes (stronger form)	none
$L = BPL$	near $L^{(3/2)}$ for 25 years	complete	yes (definitions locked)	none
Plane $\chi \geq 6$	$\chi \geq 5$ (2018)	+1, new method	yes (two-line statement)	none
Thompson F	wrong proofs in both directions	resolved	yes	none
NS blowup	forced Euler precursors	alternatives (C)/(D)	structure matches	Clay has not accepted; disputed
Vlasov–Maxwell	40 years of restricted results	complete, one species	yes	none
Mézard–Parisi	upper bound only	equality, even-arity class	yes (narrower than title)	none

15.1 Five patterns

1. **Short statements, long proofs:** plane coloring has a two-line statement and a 30k-line proof; quasi-RH a one-line statement and a 487k-line proof. This is the ideal "humans review statements, machines check proofs" split — and it makes human reviewing bandwidth the new bottleneck.

2. **Unusually short papers:** 13 pages for 9/4 matrix multiplication and for Thompson's group. Either the methods really are surprisingly lean, or much of the detail lives in Lean rather than in prose for humans — Gómez-Serrano said the NS paper is "not written for humans."
 3. **Borrowed tools across fields:** Thompson's group via Lipschitz maps from infinite-dimensional geometry; plane coloring via ergodic theory; matrix multiplication via the asymptotic spectrum rather than the dominant laser method of 40 years. This may be a real AI advantage: no human-expert "field inertia."
 4. **The gap between "satisfying the statement" and "answering the real question":** NS with freely chosen forcing; 376 putting computation in the forcing; the plane-coloring footnote about a formalized false hypothesis; Mézard–Parisi covering only even arity. Each is "spec vs intent."
 5. **Selection effects:** results picked from 4,000 problems vs 3% on pre-registered problems (§2.3). Until OpenAI publishes its failure list, "what is this model's real ability on open problems?" cannot be answered.
-

AI AND AI SAFETY PERSPECTIVE

16. What this means for AI

16.1 Better at math ≠ smarter across the board – but a leading indicator

Mathematics has a unique advantage for AI: **answers can be machine-checked** (Lean), which lets reinforcement learning iterate massively — try, get feedback, improve. Chollet's question goes to the heart of it: does progress in these "verifiable domains" transfer to unverifiable ones? A reasonable reading: **any intellectual work that can be automatically checked will fall at a similar pace next**; mathematics is just the first.

16.2 The trust bottleneck moves from "proof" to "specification"

Once there is a Lean proof, the question shifts from "is the proof right?" to "is the statement encoded right?" This is isomorphic to a core problem in alignment research: a system can perfectly optimize the objective you wrote down, and what you wrote down may not be what you want. The line-by-line reading of Lean statements done throughout this deck is a form of **spec review** — an increasingly important and scarce human job.

16.3 Knowledge that can be verified but not understood

In the words of the IAS advisory group: *"It is now the case that AI can output mathematical arguments in situations without the human who prompted it being able to understand the arguments, verify them, or take responsibility for them."* [R] Mathematics is where this situation appears first and most cleanly (because verification is perfect). Generalized to science, humans may move from "understanders" to "checkers." Tao's Math 2.0 — putting exposition, community building and opening new directions on a par with problem-solving — is at bottom an answer to "what do humans do in this new division of labor?"

16.4 Two sides for safety research

Good news

- **Verifying is easier than discovering:** if future AI can give formal proofs about its own behavior ("this action will not do X"), humans need not understand all of its capabilities, only check the proofs — the core idea behind the Guaranteed Safe AI agenda, which becomes more feasible as mathematical AI improves.
- This Lean library is itself the first real sample of "large-scale AI output + machine verification," and its failure modes can be studied.

Bad news

- The hardest parts of alignment ("what does this AI really want?", "is the outcome good for humans?") cannot be written in Lean. Verifiable domains racing ahead while unverifiable ones lag is itself a risk.
- The verifiable parts of AI R&D (kernel optimization, proofs, benchmarks) will be the first to be automated at scale — the "AI improves AI" loop is most likely to start there.

16.5 Governance: a precedent of unilateral dumping

The community had begun forming norms through the IAS advisory group (disclose model name, prompts, reasoning chains, compute time and cost, how problems were chosen, failure rates on comparable problems). OpenAI followed some of them, then released 700+ papers at once. AHM's phrase "a demonstration of power" is pointed, but it names a structural problem: **a frontier company unilaterally decides how a capability enters the public domain, and the affected community can only react afterwards.**

17. Looking ahead: extrapolate the trend, not the intuition

Part I covered what has already happened. Part II takes on a bigger question: **at the rate AI is improving at mathematics and programming, when, and in what order, will several lines of AI-safety theory (debate, singular learning theory, Guaranteed Safe AI, neurosymbolic systems) be unlocked?**

WHAT "REALISTIC" MEANS HERE

Human intuition extrapolates linearly, but the three curves below are exponential. So a trend-based projection often feels "too fast." The rule for this part: **don't be conservative just because something is counterintuitive, and don't make things up.** Every date names the curve it depends on, the assumptions used, and the signals that would falsify it. Tags: [data] sourced; [model] computed from explicit assumptions; [judgment] my inference.

17.1 Curve ①: how long a task AI can complete on its own

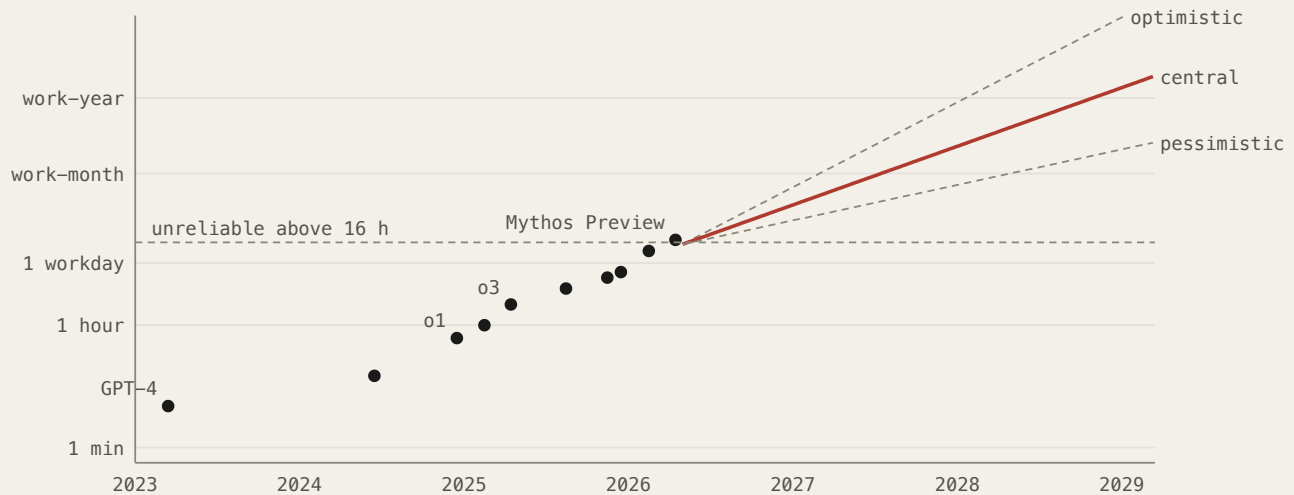
What it measures: METR gives AI a set of software and research tasks, first measuring how long each takes a human expert, then finds the task length at which the AI succeeds 50% of the time. That length, the time horizon, compresses "how capable is the AI" into one number you can plot.

RELEASE	MODEL	TIME HORIZON (P50)
2019-02	GPT-2	0.1 minutes
2023-03	GPT-4	4 minutes
2024-12	o1	39 minutes
2025-04	o3	2.0 hours
2025-08	GPT-5	3.4 hours
2026-02	Claude Opus 4.6	12 hours (CI 5.3–61)
2026-04	Claude Mythos Preview	17.4 hours (CI 8.5–55)
2026-06	GPT-5.6 Sol	cheating counted as failure: 11.3 hours; other treatments: 71–270+ hours

Source: METR TH1.1 raw data benchmark_results_1_1.yaml (updated 2026-05-08); GPT-5.6 Sol from METR's 2026-06-26 report. [data]

Doubling time: fitting from 2023, the horizon doubles every **129 days** (95% CI 104–158 days); using only post-2024 data, about 89 days; all data since 2019, about 188 days. [data] A 129-day doubling is $2^{(365/129)} \approx \mathbf{\times 7 \text{ per year}}$. That is the root of the counterintuitive feel: linear intuition expects next year to be "a bit better"; the trend says "seven times better."

Curve ①: task length AI completes alone (METR p50, log scale)



Black dots: state-of-the-art models in the raw METR TH1.1 data (from 2023; GPT-2 in 2019 at 0.1 minutes is off-chart). Red: central scenario, anchored at ~15 hours in 2026-05 and doubling every 129 days (METR fit from 2023, 95% CI 104–158 days), i.e. roughly $\times 7$ per year. Dashed: optimistic (89 days, the post-2024 pace) and pessimistic (213 days, ~7 months). Each gridline step on the vertical axis is 10 $\times$. Curves computed by the build script.

SCENARIO	DOUBLING	END 2026	MID 2027	END 2027	END 2028
Optimistic	89 days	~100 h	~410 h	~1,700 h (10 work-months)	~30,000 h
Central	129 days	~56 h (1+ work-week)	~150 h (≈ 1 work-month)	~400 h (≈ 2.4 work-months)	~2,800 h (≈ 1.4 work-years)
Pessimistic	213 days	~33 h	~60 h	~110 h	~360 h (≈ 2 work-months)

[model] Anchored at ~15 hours in 2026-05 (between Opus 4.6 and Mythos Preview), pure exponential at a fixed doubling time, computed by the build script.

Two weaknesses of this curve (keep both in mind): (1) METR says its current task suite is unreliable above 16 hours, so post-2027 numbers need a new long-task suite (TH 2.0, not yet released) to be tested; (2) the GPT-5.6 Sol result shows that whether cheating counts as success can move the number by about 4.5 doublings — roughly 1.5 years of trend.

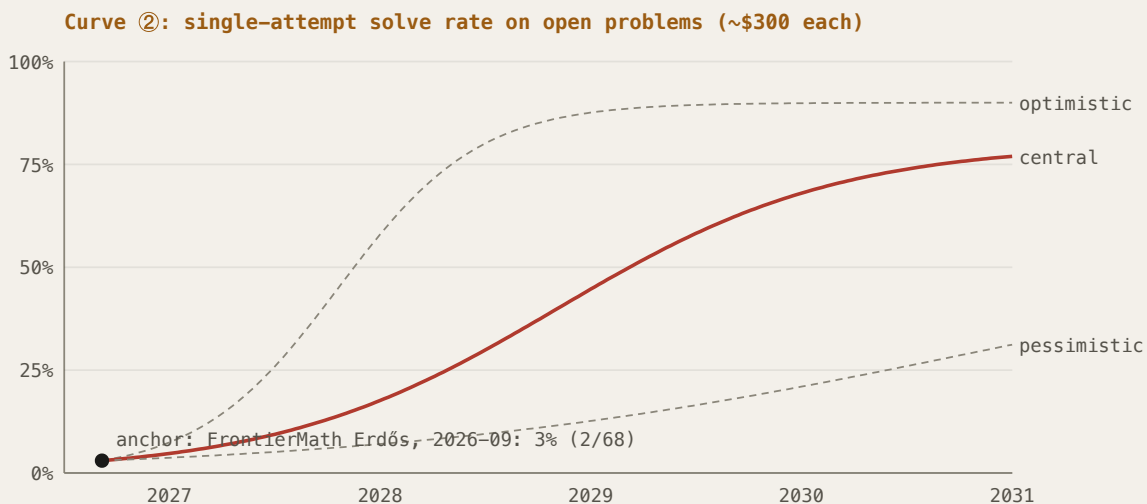
[judgment] Beyond about one work-month, success depends more and more on whether the task is clearly specified rather than on execution ability.

17.2 Curve ②: how many genuinely open problems AI can solve

Why a second curve: competitions and benchmarks are saturated and can no longer be extrapolated. [data]

- IMO 2024 silver (28/42) → 2025 gold (35/42) → 2026 perfect score (42/42, two systems with official grading).
- PutnamBench (Lean): about 7% in 2025-04 → 672/672 in 2026-08; in 2026-09 the full set was solved for \$111.
- FrontierMath Tier 4 went from 19% (2025-11) to 98% (2026-09).

Over the same period, on Epoch's **FrontierMath Erdős** (68 Erdős problems pre-registered as open, Lean proofs required, \$300 budget each), the strongest model, GPT-6 Astra, solved only **2/68 = 3%**; all others scored 0. With unlimited budget it reached 5/68 (about 7%) at a cost above \$220,000. [data] **The gap between 98% on benchmarks and 3% on open problems is the single most important uncertainty in the whole extrapolation.**



Scenario parameters: optimistic +3.0 logit/yr, cap 90%; central +1.5, cap 80%; pessimistic +0.7, cap 60%. Model: solve rate = $C \times \text{sigmoid}(a + s \cdot t)$, where C caps the fraction of problems that are extremely hard or have no short proof; anchored at the 3% that GPT-6 Astra scored on Epoch FrontierMath Erdős (68 pre-registered open problems, Lean proofs required). For comparison, FrontierMath Tier 4 rose about 5.3 logits in 10 months; all three scenarios are slower because open-problem difficulty is heavy-tailed. Curves computed by the build script.

SCENARIO	SLOPE · CAP	END 2026	MID 2027	END 2027	END 2028	END 2030
Optimistic	+3.0/yr · 90%	~7%	~26%	~58%	~88%	~90%
Central	+1.5/yr · 80%	~5%	~9%	~18%	~45%	~77%
Pessimistic	+0.7/yr · 60%	~4%	~5%	~7%	~13%	~31%

Why the central case may be conservative: (1) a 10× larger budget already lifts 3% to about 7% in 2026, and inference prices fall more than 10× per year (Epoch: 9–900× per year at fixed capability, median 50×) [data]; (2) OpenAI kept 372 families out of ~4,000 problems (about 9%), so a possibly stronger unreleased model, under "many attempts plus post-hoc selection," is already well above 3%. **Why it may be optimistic:** on a fixed problem set the remaining problems get harder; single Millennium-level problems are outside the model's scope.

17.3 Curve ③: the cost of formal proof

- **2025-09:** Math Inc's Gauss formalized the strong prime number theorem in 3 weeks, about 25,000 lines (human experts had made partial progress over 18 months). [data]
- **2026-09:** Claude completed a full formalization of **Fermat's Last Theorem** in 11 days: 13 million lines, about 6 billion output tokens, only the three standard axioms, reviewed by Buzzard. At list price about \$300,000, i.e. **\$0.023 per line**. For comparison, the human project was budgeted at 5 years and £1 million. [data]
- **2026-10:** OpenAI's Lean library, 26 million lines (see §3).

SCENARIO (YEARLY DECLINE)	END 2026	END 2027	END 2028	A FERMAT-SCALE PROJECT, END 2028
Optimistic (÷30/yr)	\$0.008/line	\$0.0003/line	\$0.00001/line	~\$110
Central (÷10/yr)	\$0.01/line	\$0.001/line	\$0.0001/line	~\$1,400
Pessimistic (÷3/yr)	\$0.016/line	\$0.005/line	\$0.002/line	~\$23,000

Implication: in all three scenarios, **the cost of generating proofs becomes negligible by 2027–2028**. What remains is almost entirely the cost of humans checking whether statements and specs are faithful, and that throughput barely grows over time. △ Some reports put the internal cost of the Fermat formalization at about \$2,000, roughly 150× below list price; the basis is unverified.

17.4 What could break the trends

FACTOR	DIRECTION	EVIDENCE AND JUDGMENT
Full automation of AI R&D	faster	The CASP 2026 paper (co-signed by Hinton, Bengio, OpenAI's chief scientist and others) estimates $r \approx 1.39 > 1$, i.e. superexponential growth: after full automation, the pace of progress rises 10× in about 17 months. Precursor: Anthropic's share of AI R&D that is "AI-led, human-supervised" rose from 1% to 26% between 2026-03 and 2026-08; fully autonomous is still 0%. [data]
Slower growth of training compute	slower	Possibly from 4–5× to 3–4× per year (power, HBM). Mathematics and formalization mostly consume inference compute, so they are less affected. [judgment]
Data	neutral to positive	In verifiable domains the Lean kernel or a test suite is an unlimited reward signal; tens of millions of new Lean lines flow back into training. The data wall is weakest here. [judgment]
Noisy verifiers	distorts measurement	An audit found errors in 42% of FrontierMath problems; about 30% of SWE-bench Pro tasks are flawed. Metrics fail before capabilities do. [data]
Reward hacking	distorts measurement	Extrapolations of METR must report the "cheating-treatment interval."
Regulation and social response	locally slower	The Fields medalists' letter and IAS publication norms slow academic acceptance of AI results, but barely constrain capability itself.

PART II · LOOKING AHEAD · FRAMEWORK

18. The core insight: AI acceleration is very uneven, so bottlenecks move

All three curves measure the same kind of thing: **work with an automatic judge**. The Lean kernel judges whether a proof is right, tests judge whether code is right, benchmarks judge whether an answer is right. AI improves extremely fast on such work, because reinforcement learning can loop indefinitely: try → get reliable feedback → improve.

HAS AN AUTOMATIC JUDGE (AI WILL DOMINATE)	NO AUTOMATIC JUDGE (STILL MOSTLY HUMAN)
Given a theorem statement, find a proof	Decide which theorem to prove , and whether the statement captures what you actually care about
Given a spec, write code and prove it meets the spec	Write the right spec (e.g. what "harmless" means)
Given a model, compute its mathematical properties	Decide how that quantity relates to "safety"
Run experiments, sweep hyperparameters, replicate	Design the right experiment and decide what the results mean

18.1 Evidence that the bottleneck is already moving

- **Proof success rates exploded:** on CLEVER (161 HumanEval problems translated into Lean, requiring code plus a proof that it meets the spec), end-to-end success went from 1/161 in 2025-05 to about 87.5% in 2026-05 (a Claude Code agent). DafnyBench went from about 68% to 96% in a year. [data]
- **Spec quality did not keep up:** the vericoding benchmark (Tegmark's group, 12,504 specs) found that among **successfully verified** samples, about 9% of specs were too weak ("vacuous specs": e.g. a sorting spec that omits "the output is a permutation of the input," so returning an empty list passes) and about 15% had translation problems. [data]
- **Statements go unreviewed:** OpenAI's 416 Comparator challenge files are marked `review: unchecked` (§3); Fermat could be confirmed only because Mathlib already had a human-written statement to compare against. [data]
- **The IMO 2026 pipeline** added a dedicated judge after the Lean proof to check whether the formal statement covers the original problem. Statement alignment has become a problem in its own right. [data]

18.2 Splitting "needs more math" into two kinds

Saying a field "needs more mathematics" covers two very different situations:

- **Type A: the statement is clear; only the proof is missing.** Example: "Is Prover-Estimator Debate still sound without the stability assumption?" — a well-defined proposition, and with Lean it has an automatic judge. **AI will clear these in bulk around 2027.**
- **Type B: it is not even clear what to prove.** Example: "What quantity should characterize the gap between SGD-trained networks and the Bayesian posterior?" — the hard part is

finding the right statement. AI can propose conjectures and test them numerically, but no automatic judge says "this is the right question." **AI helps slowly here.**

AREA	TYPE A (MISSING PROOF)	TYPE B (MISSING STATEMENT)	ENGINEERING / EXPERIMENT	CONCEPTUAL
Debate	PE debate without stability; protocols robust to an ϵ fraction of judge errors	Which real arguments satisfy stability; how to formalize equilibria with non-zero-sum play or collusion	Can RL reach the honest equilibrium (exploration hacking); testbeds with real capability gaps	What "honest" means without ground truth; true-but-selective statements
SLT	Learning coefficients of specific architectures (symbolic computation); Langevin dynamics and the LLC	The gap between SGD and the Bayesian posterior; what finite-sample and ReLU versions should look like	Estimator robustness; scaling to 10B+ models	How λ relates to "deception" or "reward hacking"
GSAI	Proofs for software and protocols (largely automatable already)	Composition theory for sound bounds over credal sets	Tools to draft and review specs; repository-scale verification; hardware	Where physical disturbance bounds come from; "harm" in social domains cannot be written down
Neurosymbolic	(mostly engineering already)	Is there a sound check for the translation layer?	"LLM proposes, solver judges" as the default architecture	Which tasks can be decomposed this way

PART II · LOOKING AHEAD · AREA 1

19. AI safety via debate

Intuition To decide whether the number of primes between 1 and 2^{60} is odd, a human cannot count. Rules: Alice states a total, splits the range in half and states the count for each half; Bob challenges one half; they keep halving... After about 60 rounds a single number remains, and the judge checks whether it is prime. If Alice lies, **at least one half is wrong**, and Bob can always follow the error down. So a single cheap check by the judge indirectly verifies an exponential computation.

Theorem	Irving–Christiano–Amodei 2018: with optimal play, debate extends the problems a judge can decide from NP ("give me a certificate and I'll check it") to PSPACE . Debate is zero-sum, and for a true claim honesty is a winning strategy, so the equilibrium is honest — that is what "the Nash equilibrium of debate is honesty" means. The authors call it an analogy and do not expect practical ML to reach all of PSPACE. [data]
Hidden assumptions	(a) strong enough play; (b) a judge who is reliable on each small step; (c) problems that decompose recursively with checkable steps; (d) zero-sum, no collusion; (e) training that actually converges to the equilibrium.
2023 refinement	Doubly-efficient debate (Brown–Cohen, Irving, Piliouras): the honest side needs only polynomially many steps to beat a liar allowed exponentially many. "The honest side is no stronger than its opponent" is enough.

19.1 Where it is stuck

1. **Obfuscated arguments (2020)**: a liar decomposes a simple problem into many subproblems, *one of which contains an error*, but **neither side can find which** (finding the error is computationally hard). The 2018 theorem assumed the honest side can locate errors. **Prover-Estimator Debate** (2025, 2506.13609) partly fixes this: the estimator no longer picks what to attack but assigns probabilities to each subclaim; it needs a **stability assumption** — the conclusion does not flip under small changes in those probabilities. **Whether real arguments are stable, nobody knows.**
2. **Learnability of the equilibrium**: a 2026-04 paper (2604.28182, with GDM researchers) shows models can be trained to resist RL exploration, and frontier models that know they are in training explicitly reason about "suppressing exploration." [data]
3. **Systematic judge errors**: humans err consistently on some sub-distributions. Irving proposed (2025-05) making arguments robust to an ϵ fraction of errors; whether a version exists that liars cannot exploit remains open.
4. **True but selective statements** (2609.29189): the verdict is right, but selection and framing serve a hidden goal. Debate's correctness guarantee does not reach this layer.
5. **Collusion**: when both debaters are copies of one model, what keeps the game zero-sum? 2603.05293 finds that with identical training corpora debate collapses to RLAIIF.

19.2 Evidence

WORK	SETTING	RESULT
Michael et al. 2023 (2311.08702)	human debaters, judge without the source text	judge accuracy 84% with debate vs 74% with a single consultant
Khan et al. 2024 (ICML best paper)	LLM debaters	non-expert LLM judges 48% → 76%; humans 60% → 88%
Kenton et al. 2024 (GDM)	math, code, logic, multimodal	debate > consultancy everywhere; versus the judge answering directly, better only on information-asymmetry tasks
Engels...Tegmark 2025 (2504.18530)	oversight success vs capability gap	nested-oversight success at a 400-Elo gap: debate 51.7% (Mafia 13.5%, backdoored code 10.0%)
Kenton et al. 2026-08 (GDM, 2608.17776)	math RL with debate as the reward signal	the RLAIIF baseline quickly learns to exploit the judge; debate training keeps the judge effective, recovering 45% of the performance gap

[judgment] The evidence supports debate as **a reward signal that is harder to hack**; it does not yet support "debate lets humans reliably oversee superhuman systems." Every experiment's capability gap is far smaller than the theory imagines.

19.3 What AI mathematics and coding can do

- **Type A, now:** formalize and prove the theorems about debate protocols. Irving said in a 2026 80,000 Hours interview: "theory is easier to automate than empirics, because proofs give a verifiable reward." This is the core bet of Resolution, where he is chief scientist. [data]
- **A new opportunity: debate experiments with real capability gaps, built on formal mathematics.** AI's formal-math ability already far exceeds most humans', so "a human judge overseeing an AI's mathematical argument" is a natural superhuman-oversight experiment, with Lean providing perfect ground truth — much closer to the real problem than QuALITY-style hidden information. [judgment]
- **A counterintuitive point:** where things can be formalized, Lean is already a perfect judge and debate is unnecessary. **Debate's value lies precisely outside the formalizable boundary**, which is exactly where AI helps least.
- **What it cannot help with:** defining "honesty" for value judgments; selective statements.

20. Singular learning theory (SLT)

Classical theory

Classical statistics assumes the best parameter is a single point surrounded by a bowl. Then the "effective number of parameters" is the true count d , and generalization error is about $d/(2n)$.

Toy example

Model $y = a \cdot b \cdot x$ with data from $y = 0$. The loss is about $(ab)^2$, which is optimal whenever $a = 0$ or $b = 0$. So the optimum is a **cross**, not a point; at the crossing the Hessian is all zeros and classical approximations fail. This is called **singular**.

Counting effective parameters

Look at how the volume of "nearly optimal" parameters $V(\epsilon)$ shrinks as $\epsilon \rightarrow 0$: $V \approx \epsilon^\lambda$. One-parameter bowl a^2 : $\lambda = 1/2$; two-parameter bowl $a^2 + b^2$: $\lambda = 1$; singular $a^2 b^2$: $\lambda = 1/2$. Two nominal parameters behave statistically like one. λ is the **learning coefficient** (RLCT).

Theorem

Watanabe (2009, 2018) proved that for Bayesian inference the free energy $\approx n \cdot S + \lambda \cdot \log n$ and the generalization error $\approx \lambda/n$. Neural networks are almost always singular, so λ , not the parameter count, governs generalization. The proof uses Hironaka's resolution of singularities. [data]

20.1 Proven vs conjectured

- **Proven:** the asymptotic formulas above (for real-analytic models on compact parameter spaces, etc.); exact λ for a few architectures such as deep linear networks (Aoyagi).
- **Unproven:** that SGD-trained networks match the local structure of the Bayesian posterior; that training "stages" correspond to Bayesian phase transitions; that λ describes finite-sample, ReLU, non-realizable real LLMs.

20.2 Evidence

WORK	SCALE	RESULT
Lau et al. (2308.12108)	deep linear networks, etc.	the local learning coefficient (LLC) estimator recovers known exact λ
Hoogland et al. (2402.02364)	~3M-parameter transformer	plateaus and turns in the LLC mark discrete developmental stages of in-context learning
Kreer et al. (ICLR 2026)	billions of parameters	Hessian-free data attribution at SOTA
Spectroscopy at Scale (2026-04)	Pythia-1.4B	57,236 structural clusters, 4,800 H200-hours
Patterning (2601.13548)	small models	reweighting data along susceptibility directions can speed up or delay the emergence of induction circuits

[data] The largest scale is about 1–3B; **there are no frontier-model results, and no case where SLT found a safety problem that other methods could not.** The LLC is estimated by SGLD sampling, which the authors concede loses unbiasedness and is hyperparameter-sensitive.

20.3 What AI can do

- **Now:** (a) symbolic computation (Newton polyhedra, blow-up algorithms) to produce exact λ for many small architectures, giving estimators ground truth; (b) agents that sweep hyperparameters and replicate across seeds, fixing the estimator-robustness pain point.
- **Medium term:** running LLC/susceptibility pipelines on large models is mostly compute-limited; human effort can be driven very low.
- **Hard:** Lean formalization (Mathlib lacks Hironaka's theorem; a huge undertaking); a theory of the SGD–Bayes gap (type B).

The decisive experiment: on a model organism with an implanted backdoor or reward hacking, can λ or a susceptibility give a warning **before the behavior shows**? This decides whether SLT becomes a standard tool for monitoring training or remains a scientific tool for understanding learning dynamics. [judgment] Agents will bring the cost of this experiment down to a few weeks for a small team.

21. Guaranteed Safe AI and neurosymbolic systems

The GSAI idea (Dalrymple, Skalse, Bengio, Russell, Tegmark et al., arXiv 2405.06624): instead of understanding an AI's internals, require each output to carry a **certificate** proving it satisfies a **safety specification** in a **world model**, checked by a trusted **verifier**.

Example A: verifying a sort function

Spec: "the output is sorted and is a permutation of the input." The world model is the programming language's semantics. Lean/Dafny proves it for all inputs. **Pitfall**: drop "is a permutation of the input" and returning an empty list also passes.

Example B: verifying a braking controller

World model: position and velocity update equations plus "road disturbance at most 0.1." Spec: never hit the obstacle. The verifier does reachability analysis. **Pitfall**: the certificate holds only inside the model; on ice the disturbance exceeds 0.1 and the guarantee fails, so runtime monitoring and a fallback controller are needed.

Replace example B with "a chatbot never gives harmful advice": the world model becomes "human society" and the spec becomes "harm," and neither can be written down. That is GSAI's hardest end.

21.1 The flagship institution has pivoted (ARIA Safeguarded AI) [data]

DATE	EVENT
2024-02	Thesis v1: TA1 mathematical framework / TA2 ML (a planned £18m new organization) / TA3 applications; £59m total
2025-11	davidad halts TA2 Phase 2: "every frontier model has more capability than I expected"; money moves to TA1
2026-02	Applications narrowed to cybersecurity; cyber-physical Phase 2 cancelled
2026-04	Nora Ammann becomes programme director; davidad becomes technical advisor
2026-05	Thesis v2: keeps only "mathematical-assurance tooling + cybersecurity applications"; reasons given are that frontier progress was faster than expected and that timely international consensus on a gatekeeper workflow has become less likely, hence a turn to "accelerating improvements to societal resilience"

[judgment] The original thesis was "AI capability first, safety certificates as the gate"; v2 effectively retreats to "use AI to make formal methods cheap and harden critical software." The gatekeeper is now a long-term option only.

21.2 AI is taking over the verifier leg [data]

BENCHMARK / CASE	EARLY	LATEST
DafnyBench (782 programs)	~68% (2024)	92.7%–96% (2025–2026)
CLEVER (161 problems, Lean, end-to-end)	1/161 (2025-05)	~87.5% (2026-05)
VERINA (189 problems, Lean)	o3 proof success 4.9% (2025-05)	an 8B specialized model averages 62% on three Lean benchmarks (2026-03)
seL4 (8,700 lines of C)	by hand: ~20 person-years, ~\$350/line (2009)	a 7B model proves 51.7% of the lemmas in seL4's important theories (2026, statements given)

Pattern: the same benchmark goes from single digits to 80–90% in about a year, the same shape as HumanEval in 2023. Dafny is easier than Lean because it hands part of the reasoning to an SMT solver and the model only writes hints — which is itself neurosymbolic.

21.3 Order of arrival [judgment]

DOMAIN	WORLD MODEL	SPEC	EXPECTED
Crypto libraries, protocols, kernels, C→Rust equivalence	language semantics, exists	often exists (RFCs, reference implementations)	now to 2027
Hardware, RTL, firmware, distributed protocols	ISA, memory model; engineering-limited	writable, needs human review	2027–2029
Control systems with reliable physics (grid, braking)	physical laws + disturbance bounds; math-limited	safe sets; conservative but writable	later; "finite-horizon certificate + runtime monitoring"
Open-world social and biological domains	no auditable model	"harm" cannot be written down	last, perhaps only weak guarantees forever

A counterintuitive consequence: AI makes finding vulnerabilities cheaper, but also makes proving their absence cheaper. **In software, the offense–defense balance may tilt toward defense** — provided defenders actually do formal verification. Biology and social domains lack this symmetry, because no spec can be written.

21.4 Neurosymbolic systems

A neural network "guesses" (proof steps, spec drafts, programs, world-model code) and a symbolic system "judges" (Lean kernel, SMT solver, reachability analysis). The judge is reliable; the guesser can be a black box — the same structure as GSAI's "black-box policy, trusted verifier."

- **Status:** it is already the de facto default architecture in every verifiable domain. Programmatic world models (PoE-World, NeurIPS 2025: an LLM writes 4,000+ lines of Python to model the Montezuma game world and generalizes zero-shot to new levels) show the route works, but only in Atari-like settings. AWS's Automated Reasoning checks translate text into logic and decide it with a solver. [data]
- **The weakest link is the translation layer:** every step from natural language to formal language can produce a vacuous spec or a mis-formalization, and this layer has **no sound check** — only cross-consistency of independent translations, testing, and human review.
- **Another attack surface:** DeepSeek-Prover-V2 exploited a toolchain bug in Lean 4.9.0 to get false proofs to compile. [data] "Only the checker needs to be trusted" must itself be maintained by engineering; whether RL-trained provers systematically seek such holes is a reward-hacking question.

22. Timeline: T0 → T4

Each node has four parts: **capability** (central scenario of the three curves), **what unlocks**, **new opportunities**, and **signals that would falsify it**. Dates carry roughly ± 1 year of uncertainty: the optimistic scenario shifts everything half a year to a year earlier, the pessimistic one a year or two later.

T0 · Now (2026 Q4)

- **Capability:** AI completes about 1–2 workdays of work on its own; single-attempt open-problem solve rate about 3–5%; formalization about \$0.01–0.02 per line; competition math saturated.
- **Unlocks:** theorems about debate protocols can be handed to AI to formalize; SLT's symbolic λ computations and estimator tuning can be handed to agents; the proof step of software verification is largely automated; neurosymbolic is the default in verifiable domains.
- **Opportunities:** the most valuable work is **what cannot yet be automated** — methods for reviewing statements and specs; reward hacking against provers and verifiers; designing oversight experiments with real capability gaps.
- **Falsifier:** a new METR long-task suite shows progress stalling at 1–2 days.

T1 · Mid 2027

- **Capability** [model]: task length about 150 hours ($\approx$ one work-month); open-problem solve rate about 9%; proof costs down another order of magnitude.
- **Unlocks: type-A theory problems start falling in bulk** — debate-protocol variants, λ for specific architectures, all kinds of "remove this assumption" theorems. One theorist plus AI can do in a month the proof work a group used to do in a year. SLT estimator robustness is essentially solved. Crypto and protocol libraries ship with proofs by default.
- **Opportunities:** the "AI-assisted theorist" becomes a role: you pose statements; AI proves them or finds counterexamples. The scarcest skill is **knowing which statement is worth proving**. Superhuman-oversight experiments built on formal math become practical.
- **Falsifiers:** the next round of FrontierMath Erdős is still below 5%; Anthropic's "AI-led, human-supervised" R&D share stalls around 30%.

T2 · End 2027

- **Capability** [model]: task length about 400 hours (2–3 work-months); open-problem solve rate about 18%; a Fermat-scale formalization costs a few thousand dollars.
- **Unlocks**: the idealized-model theory of debate largely converges; what remains — "do real arguments satisfy stability?", "does training reach the honest equilibrium?" — is empirical and ML work. SLT's decisive early-warning experiment becomes feasible. Hardware and RTL verification begins to scale.
- **Opportunities**: "**auditing statements and specs**" becomes a field: how to judge quickly whether a formal statement is faithful, how to design cross-consistency checks, how to measure human reviewers' throughput. Almost nobody works on this today.
- **Falsifier**: by end 2027 there is still no field-level important problem solved mainly by AI and confirmed by human experts, meaning curve ②'s cap was set too high.

T3 · 2028

- **Capability** [model]: task length about 2,800 hours (more than a work-year); open-problem solve rate about 45%; proof costs negligible.
- **Unlocks**: **most type-A problems are solved or proven impossible**. In all four areas only type-B and conceptual problems remain. The value of debate and SLT depends on one thing: whether they give guarantees outside the formalizable boundary. "Critical software ships with proofs" could become an industry norm (if regulation or insurance pushes it).
- **Opportunities**: **conceptual work has the highest return**: people who can turn "deception," "reward hacking" or "harm" into measurable, provable statements are the bottleneck itself. If AI R&D is highly automated, AI-safety theory itself is automated at scale, and **can humans review fast enough?** becomes the central question.
- **Falsifier**: task length stalls at a few hundred hours (success depends more and more on the spec), turning curve ① into an S-curve.

T4 · 2029–2030

- **The scenarios diverge**:
 - **Full automation of AI R&D** (the CASP scenario): every date compresses; theory, proofs and experiments are almost fully automated, and the human role concentrates on setting goals, reviewing statements and deciding deployment.
 - **A hard bottleneck appears** (compute, data, or spec review): progress returns to exponential or slower; type-A problems are solved while type-B and conceptual

problems move slowly.

- **True in both:** safety guarantees for software and protocols are far stronger than today, and the verification asymmetry favors defense; guarantees in the open world (biology, society) remain weak and rely mainly on monitoring, access control and governance; **who reviews the statements and specs** is the ceiling of the whole system.

22.1 Where this projection is most likely wrong

1. **Curve ①'s measurement is breaking down:** unreliable above 16 hours, and cheating treatment can shift it by 4.5 doublings. Post-2027 numbers rest mainly on "other signals are rising in step."
2. **Curve ② has a single anchor** (3%, 68 problems), and the slope is a conservative choice with wide error.
3. **The type-A/type-B boundary will move:** AI may become good at "proposing good statements" sooner than expected. There is no evidence this has happened yet — OpenAI's 719 papers mostly solve existing problems and propose almost no new concepts (compare Tao's 21 dimensions of "good mathematics"). If it happens, every node moves earlier and the human role shrinks further.
4. **Variables left out:** pacing or pause agreements, major safety incidents, geopolitics. These can only slow the timeline.

22.2 For researchers: what is worth doing now

Rule: **don't do by hand what AI will do within a year, unless it is infrastructure or helps you judge "what to ask."**

OPPORTUNITY	WHY NOW
Reward hacking against provers and verifiers	A prover has already exploited a Lean toolchain hole; most valuable to study before AI proving is deployed at scale
Auditing the faithfulness of statements and specs	The shared bottleneck of every area from T1 to T3; the line-by-line reading of Lean statements in this deck is an early version
Superhuman-oversight experiments on formal math	Existing experiments manufacture their capability gaps; formal math gives a real gap plus perfect ground truth
SLT "early warning" experiment	The experiment that decides SLT's safety value; nobody has run it
Manipulation-resistance of AI R&D automation indicators	CASP's policy proposals rely on these indicators, and reward hacking will inflate them

Key numbers in this part are sourced inline. The four underlying research drafts (debate, SLT, GSAI and neurosymbolic, trajectory data; each item tagged [P]/[R]/△ with links) are kept in the author's research notes and are not published with this deck.

REACTIONS · IN THEIR WORDS

23. What people said

WHO	QUOTE	SOURCE
Levent Alpöge (Anthropic)	"It's obviously the most significant moment in mathematical history." (later adding there would be "sad stories" of people being scooped)	X [R]
Terence Tao	"Problems are being solved autonomously by AI prompters who have no interest in the broader field itself"; proposes Math 1.0 → 2.0	Fortune [R]
Dan Litt (Toronto)	"My view is that this is great for mathematics," while worrying that an "AI has solved math" perception could discourage young people	Fortune [R]
Tristan Buckmaster (NYU)	"There's likely to be a bunch of results where they take someone's work and then take it to completion"	Fortune [R]
Scott Aaronson	Post "The Mathocalypse"; on L = BPL: "its truth was never in serious doubt"	blog [R]
Lance Fortnow	"if they hold up, we've seen more progress in TCS in the last 24 hours than in the previous three decades combined"	Computational Complexity blog [R]
Mark Braverman	"Math by press release is not that healthy for math."	Quanta [R]
François Chollet	Asks whether gains in RLVR-friendly domains generalize, or whether non-verifiable domains stay bottlenecked on human data	X (paraphrase) [R]
AHM	"Releasing over 700 files at once is not a demonstration of scholarship, but a demonstration of power."; urges mathematicians to stop working with OpenAI	reposted on Tao's blog (Tao notes he does not endorse it) [P]
Álvaro Lozano-Robledo	"Keep calm and carry on studying math."; LLMs "may work within the confines of the convex hull of ideas that are currently available in the literature"	guest post on Tao's blog [P]
Andrew Sutherland (MIT)	"We should ask for receipts."	via Scientific American [R]

Note: "The Future of Mathematics" on Tao's blog (10-05) is a guest post by Jeremy Avigad, not by Tao; some coverage conflated the two.

NEXT STEPS

24. Open questions and how to keep learning

24.1 What to watch

1. Will OpenAI publish the list of ~4,000 problems and the failures? (This decides whether "what is the real capability?" can be answered.)
2. How many errors like the Weil-class paper remain in the ~58% that is unformalized? How will the withdrawal rate evolve?
3. Who reviews the faithfulness of the Lean statements? When does `review: unchecked` change? Who audits the `lana-agents` dependencies and the large patches?
4. When will the first human domain expert give an independent "read and confirmed" verdict on a headline result (quasi-RH, UGC, Thompson F)?
5. What priority norms should govern "completing someone else's work" (the Buckmaster episode)?
6. Chromatic number of the plane: what does a finite unit-distance graph needing 6 colors look like?

24.2 Learning path (starting from linear algebra + calculus)

STAGE	MATERIAL	WHY
1 · What math is	Gowers, <i>Mathematics: A Very Short Introduction</i>	150 pages by a Fields medalist: abstraction, proof, what "hard" means
1 · Roots of this debate	Thurston, On Proof and Progress in Mathematics; Gowers, The Two Cultures of Mathematics; Tao, What is good mathematics?	after these, "Math 1.0 $\rightarrow$ 2.0" makes complete sense
2 · Touch a proof	Natural Number Game (Kevin Buzzard, free online)	prove commutativity of addition from scratch in Lean — then §3 makes full sense
2 · Learn proofs systematically	MIT 6.042 Mathematics for Computer Science (OCW + free textbook)	designed for CS backgrounds: proof methods, discrete math, probability
3 · Intuition	3Blue1Brown: <i>Essence of linear algebra</i> ; visualizing the Riemann zeta function	the figure in §4 is a static version of that episode
3 · Following the news	Quanta Magazine (math); <i>The Princeton Companion to Mathematics</i> (as a reference)	every piece explains why a problem is hard and where people were stuck
4 · ML-adjacent	Bubeck, <i>Convex Optimization: Algorithms and Complexity</i> ; Prince, <i>Understanding Deep Learning</i> (free)	convergence proofs for optimization; intro to deep-learning theory

25. Glossary and sources

25.1 Glossary

TERM	IN ONE LINE
Lean / Mathlib	a proof assistant / its community math library
sorry / axiom	in Lean: "leave this step empty" / "assume this is true" — both weaken a proof
Comparator	tool checking "the theorem proven = the challenge statement, using only allowed axioms"
zero-free region	a region where ζ or an L-function has no zeros
irrationality exponent μ	how "unusually well" a number can be approximated by fractions
ω	the optimal exponent of matrix multiplication, $2 \leq \omega \leq 3$
approximation ratio	the fraction of the optimum a polynomial-time algorithm guarantees
L / RL / BPL	log-space deterministic / one-sided randomized / two-sided randomized computation
amenable	a group admits a translation-invariant "average"
blowup	a solution of a smooth equation going to infinity in finite time
replica symmetry breaking (RSB)	Parisi's description of the hierarchical state structure of spin glasses
galactic algorithm	asymptotically faster, but only better at astronomical sizes

25.2 Main sources

Primary: [openai/math](#) (README, history.md, overview.pdf, CONTENTS.md, lean/, preprints/, reasoning_traces/; snapshot fd4aeeb, 2026-10-08) · [openai/NavierStokesAndEuler](#) · Clay NS official statement (Fefferman) · [leanprover/comparator](#) · independent re-checks [davegoldblatt/openai-zeta-proof-check](#), [erenciracioglu-dotcom/openai-math-9-4-lean-check](#) · Epoch FrontierMath Erdős (arXiv:2609.25050).

Coverage and commentary: Fortune 10-07 · Nature (paywalled, full text not read [△](#)) · Scientific American · Quanta 09-08 · Quanta 10-07 · NPR 09-22 · Aaronson · Kalai · Tao's blog · Latent Space.

Classic references: Strassen 1969; Coppersmith–Winograd 1990; Alman et al. arXiv:2404.16349; Ambainis–Filmus–Le Gall arXiv:1411.5414; Goemans–Williamson 1995; Khot 2002; KKMO 2007; Raghavendra 2008; Khot–Minzer–Safra 2018; Nisan 1992; Saks–Zhou 1999; Reingold 2008; Roth 1955; Zeilberger–Zudilin arXiv:1912.06345; Meiburg arXiv:2208.13356; Guth–Maynard arXiv:2405.20552; de Grey arXiv:1804.02385; Moore arXiv:1102.0747; Tao 2016 arXiv:1402.0290; Glassey–Strauss 1986; Parisi 1979; Talagrand 2006; Mézard–Parisi 2001; Ding–Sly–Sun 2015.